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Master Thesis Project

Generating personalized music playlists based on desired mood and individual listening data

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Abstract

Music listening is considered one of the most ubiquitous activities in everyday life, and one of the main reasons why people listen is to affect and regulate their mood. The vast availability and unlimited access of music has made it difficult to find relevant music that fits both the context and the preferences of the music listener. The aim of this project was to investigate the personalized relationship between music and mood using everyday technologies, focusing on how a listening experience could be adapted to the desired affect of a music listener while also taking the user's individual listening history into account. In large, the project concentrated on the possibility of using context-aware music recommendation to generate personalized playlists by focusing on the audio features and corresponding mood of the music. A web-based application was developed to act as a prototype for the study, where the application allowed users to connect to Spotify, pick a desired mood and generate a playlist. By allowing people to access music in this personalized way, a user study could be conducted in order to investigate their music listening while incorporating this recommendation tool. The findings showed that the users' found the experience to be engaging in that they could use the application as a companion to everyday tasks in addition to it being a tool for getting new, personalized music recommendations. Overall, the participants also found the generated playlists to be accurate to their music preferences and desired affective state.

Keywords: music recommendation, context-based music listening, mood regulation, affect regulation, Spotify Audio Features, music-mood classification, context-aware music recommendation

Preface

When deciding what to focus on for my master thesis, my only goal was to pick a topic which I would find exciting to write about. Luckily, I happen to study media technology, where anything is possible – even combining some of my favorite things in life (music, individualization and web development) in one project. You see, I am the kind of person that loves to create personal playlists. For everything. A playlist nerd, if you will. So naturally I wanted to come up with a way for other people to do the same. However, as this was not a one-[wo]man mission, I have some thanks to give out.

First of all, I would like to thank my supervisor, Aris Alissandrakis, for his continued support and extended tutoring hours. Without him challenging me to step outside of my own mind, this project could have taken a completely different turn. I'm especially thankful for his unwavering belief that this master thesis would somehow be completed, regardless of whether I believed it myself or not.

I would also like to thank my participants for their cooperation and, most of all, excitement in trying out my little music recommendation system.

Additionally, I want to give thanks to my reader and examiner for taking the time to read and examine my thesis.

Finally, I want to express eternal gratitude to my friends and family for their unwavering encouragement.

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1 Introduction

This chapter gives an introduction to the project, including a background to the given area as well as the motivations and contributions of the study. It also provides a description of the research problem that the study aimed to investigate.

1.1 Background

Listening and reacting to music is a universal act, something that most humans experience on a daily basis, regardless of groups and cultures and in both passive and active ways (Agres et al., 2021; Chai et al, 2020). A common conception of music is to see it as a background soundtrack for daily life, and many consider it as one of the most common companions for both leisure and work activities (Siles et al., 2019; Park et al., 2019; Helmholz et al., 2017).

Another increasingly more frequent use of music is in the form of interventions that are aimed at increasing well-being (Siles et al., 2019; Raglio, 2020). The International Federation of the Phonographic Industry [IFPI]'s (2022) yearly report on worldwide music consumption highlights this by stating that music continues to play a significant role in supporting physical and psychological well-being, with 69% of respondents finding music to be important for their mental health. Similarly, Bonde et al. (2022) point to music listening as a health-promoting activity when incorporated into daily life.

One reasoning behind why music stands out as a tool for "caring for the self" is its ability to reduce or manage measures of pain, anxiety and stress, but also for its capacity to regulate affect and mood (Skånland, 2013a; Chai et al., 2020; Howlin & Rooney, 2021; Raglio, 2020; Rickard et al., 2016). Other effects of music listening include increased feelings of pleasure, changes in social attitude and aiding the emotional response. Park et al., (2019) points out music as one of the main media resources for mood regulation and finds it to be one of the main reasons for listening.

When discussing the internal responses of individuals, researchers make a clear distinction between moods and emotions, where moods last longer and are less intense than emotions, which are typically aimed at specific events (Brinker et al, 2012; Gross, 2015). In turn, affect is a phrase commonly used as a blanket term for these different responses. One of the most widely-accepted theories in affective studies is a two-dimensional model termed the circumplex model of affect, which was proposed by Russell (1980) and has since had an important part in emotion research. The theory states that all affective states can be mapped in regards to two dimensions: valence (pleasure-displeasure) and arousal (rest-activated).

For young people in particular, music is regarded as one of the most favored leisure activities and a common strategy for regulating mood (Papinczak et al., 2015; Rickard et al., 2016). In fact, young people as a demographic use their mobile devices for emotional health optimizing strategies, such as listening to music, more than any other. Kauer et al (2021) highlight that young people have an effortless relationship to electronic devices, mostly since they rely on these technologies for the majority of their daily interactions. Mobile devices can thereby be used as tools for musical interventions, as it provides the possibility of assisting young people in self-managing their well-being and mood in practice (Papinczak et al., 2015).

IFPI's (2022) music consumption report further highlights that 74% of internet users listen to music through licensed streaming services such as Spotify or Apple Music. The popularity of these streaming services derive from the vast music library that they offer, making it possible for people to listen to their favorite tracks and artists on demand, but also from the possibility of curating and listening to personalized playlists. Siles et al. (2019) highlight that people use these music streaming services and their playlist management abilities in attempts to generate, explore and respond to moods in different ways. However, one issue that arises in this unlimited music library and its corresponding listening possibilities is for individuals to find music that fits their personal context or situation (Helmholz et al., 2017).

To overcome this challenge and manage the overwhelming amount of music available, information systems can make use of context-aware music recommendation softwares, where the user's current situation or desired emotional state is taken into consideration (Hagen, 2015; Helmholz et al., 2017). However, the subjective nature of music listening also implies that individual music taste and familiarity with music plays a significant role in meeting these musical needs (Papinczak et al., 2015; Baltazar et al., 2019). Therein lies an opportunity to investigate the utilization of individually-tailored playlists based on both music preferences and emotional context.

The music streaming service Spotify provides a web API (application programming interface) which enables developers and researchers to easily access metadata and audio features of tracks (Scarratt et al., 2021). These features describe the fundamental aspects of recorded music, such as energy, loudness and tempo, in addition to computed measures that indicate how the music is used. Several researchers underline the use of Spotify's audio features to classify and categorize music according to mood and emotions, where valence and arousal stand out as two main components (Duman et al., 2022; Howlin & Rooney, 2021; Park et al, 2019).

By integrating affective research and music technologies in this way, an opportunity arises where music can be utilized in a functional way, with a specific objective, by taking structural components into account (Raglio, 2020). Music streaming services' integration with mobile technology creates an additional potential to

examine the ubiquitous role of music listening in everyday contexts (Schriewer and Bulaj, 2016). Consequently, it appears relevant to further investigate the use of music streaming services to generate personalized playlists with regard to both musical preferences and affective states.

1.2 Motivation

There is a considerable amount of research that recognizes the effectiveness of using music for regulating affect and mood, in addition to other health-beneficial contexts. Previous studies in the field have mainly focused on the use of music for managing and regulating levels of stress, pain and anxiety (Baltazar et al., 2019; Howlin & Rooney, 2021; Raglio, 2020). The majority of this existing research has laid its focus in clinical settings, disregarding the use of music as a self-care resource in everyday life. The studies that have investigated the individual responses to music in daily settings have mainly laid its focus on why people listen to certain music in particular contexts, such as for dancing (Duman et al, 2022) or for sleeping (Scarratt et al., 2021).

When regarding current technologies, there are many existing mobile applications that claim to support the facilitation of well-being, where some are focused on tracking mood and emotional state (Rickard et al., 2016). However, the empirical research behind these mobile interventions is still growing, and rarely focuses on the potential combination of health and music technologies (Bakker & Rickard, 2017; Agres et al., 2021).

Existing recommendation systems have mainly focused on considering either user preferences or their context, not a combination of both (Helmholz et al., 2017). Research studies that have made use of Spotify's audio features have mainly focused on investigating and analyzing the act of music listening in different contexts by examining the audio features corresponding to that music (Duman et al., 2022; Scarratt et al., 2021; Howlin & Rooney, 2021). Yet, there is lack of research on how these audio features can be used as basis for creating or adapting to these contexts. Therein lies an opportunity to bridge that gap and offer a developed tactic for affect regulation in everyday life, by taking both musical features and preferences into consideration.

1.3 Problem statement

The project meant to investigate the following research question:

RQ	How can a music playlist be generated, which is personalized according to a user's listening history and desired mood?

The hypotheses for the study were as follows:

H1	Users will confirm that the generated playlists matched their desired mood.
H2	Users will confirm that the application that generates such playlists is engaging.

1.4 Contributions

By investigating the use of context-aware playlist recommendation, this project mainly contributes to existing research by creating an opportunity to use music as a supportive tool for regulating affect and mood. The idea was for young people to be able to self-regulate their affective states with a tool that they already make use of in their everyday life. In addition, the project provides an example of how Spotify's Audio Features can be used in practice and in combination with the research behind affective studies. Existing research on similar interventions is not wide, with applications and projects rarely focusing on the combination of music and mood. As such, this project could contribute to research on using music technology as a stimulator of affect and mood, and in turn investigate the relationship between young listeners' well-being and the role that music plays in their lives.

1.5 Target groups

The possible target groups that might be interested in taking part in this thesis come from two different areas: researchers or developers, or a combination of both. Researchers that specialize in the area of mood or music might take interest in how the project combines the two areas in an everyday context. On the other hand, it might interest professional developers looking to work with Spotifys API or more specifically use their audio features. The use of Spotify's Audio Features might also be of interest to other researchers looking to classify or categorize music by using these parameters, specifically if their interest lies in the context of affect and mood.

1.6 Report structure

The report has the following structure: Section 1 provides background and motivation for the problem at hand, and presents the research questions and hypotheses of the study. Section 2 presents existing research that is relevant in the area of music listening, affect regulation and music recommendation. Section 3 presents the chosen methodology both in terms of practical approaches and eventual limitations. In Section 4, the implementation of the prototype is described in terms of its major functionalities and how the playlist generation was constructed for the user. Section 5 provides the results and analysis of the collected data. Section 6 then discusses that data in relation to the given circumstances and existing research. Finally, Section 7 provides conclusions for the findings, in addition to discussing future work.

2 Related Works

This chapter presents and classifies the existing research relevant for the project.

2.1 Music listening in everyday life

2.1.1 Why do people listen to music?

People listen to music for a majority of different reasons and in a variety of different situations, and music can thereby offer a range of different functions in everyday life (Duman et al, 2022; Getz et al., 2014). Some people listen to music for entertainment, relaxation or to reduce stress, others listen in order to feel connected, to manage and create relationships or to achieve a certain ambience, while some simply use music to accompany them in their daily activities.

The research behind music listening is broad, with numerous proposed functions that aim to explain and categorize the reasonings behind this ambiguous human behavior. Duman et al. (2022) points to factors such as personal characteristics, routines and age to explain people's musical preferences and their reasons for listening. In another study that investigated music-listening contexts and their motivations, findings showed three main functions: to achieve self-awareness, to express social relatedness and to regulate arousal and mood (Schäfer et al., 2013). An additional study found that motivations for music listening can be divided into three different categories: to regulate emotions, to use in a cognitive manner and for background use, where music serves as an enjoyable companion while doing something else (Getz et al., 2014).

Another area of research explains people's music listening motives by pointing to the uses and gratification theory (Putzke et al, 2014; North and Hargreaves, 2008, p.226). The theory claims that people choose to consume specific media for the reason that it helps them to achieve and satisfy particular goals and needs, where one such need is mood-management and another is forming a personal identity.

Consequently, one of the most common themes that emerges among these reasons for listening is the enhancement and regulation of moods and emotions (Getz et al., 2014). In addition, Goethem and Sloboda (2011) highlight that music often fulfills an emotion-related function in everyday life, e.g. for energization or relaxation, to articulate emotions, or to recall certain memories. Mood-related experiences in general have demonstrated to be one of the primary reasons for consuming music and participating in musical activities (Saarikallio & Erikkilä, 2007).

2.1.2 Young people

Music is a particularly favored and frequent activity among young people, as young adults and adolescents are reported to be the biggest consumers of music with up to 18 hours of intentional music listening per week (Dingle et al., 2016). In a study by Papinczak et al. (2015) where young people's use of music was investigated, findings showed that music played a crucial part in their everyday activities, spanning a wide range of contexts. Their study mainly highlighted young people's essential use of music to enhance or restore well-being in daily life. Further, the study identified emotional management as one of the most common motivations for young people's interest in music, particularly when the participants experienced undesirable moods and levels of arousal. The findings showed that young people knew which music to play in different contexts in order to get into desired mood, achieve a certain ambiance or get aid in everyday tasks.

Another important finding that emerged in Papinczak et al.'s (2015) study was young people's use of music to amplify their emotions rather than alter them. For difficult emotions in particular, music served as a tool for listeners to immerse themselves in the negativity and fully experience and process those emotions before moving on to more upbeat music. Similarly, North and Hargreaves (2008, p.229) highlight the fact that young people are not passive music listeners, but that they actively seek out music that addresses challenging themes to better understand and handle the situations unfolding in their own lives.

2.1.3 Health potentials for music listening

That there is a relationship between people, music and well-being is not a new concept, and the idea that music can be thought of as salutary has been known and practiced throughout history (Skånland, 2013b). This combination is a field that encompasses multiple different domains, and can be perceived as a continuum between music in therapeutic settings and music as self-care in individuals' personal lives.

A study conducted by Bonde, Ekholm and Juel (2022) researched the public health potential of incorporating music in daily life. The findings showed a distinct correlation between daily incorporations of music and a higher health status, where 77% of informants considered music to be a factor that promotes health. In addition, a majority of informants reported the use of music to manage both their external and internal processes.

Additional research shows that music listening has been found to be an effective tool in different types of pain management, in addition to being a common method for reducing and regulating stress in everyday life (Howlin & Rooney, 2021; Baltazar et al, 2019). Another study by Raglio (2020) points out that music listening, over the last few decades, has increasingly been used in both clinical and personal settings to stabilize symptoms, alleviate levels of pain and stress, reduce

anxiety and induce pleasure. This act of listening to music as a means of care can be described as a form of music-based intervention, where the general aim is to enhance a person's well-being. Similarly, Skånland (2013b) highlights that music has the potential to contribute to an enhanced quality of life and improved mental health, whether introduced in theurapeutic or personal settings.

2.2 Affect and mood defined

Skånland (2013a) defines affect as a combination of emotions, moods and stress responses. To distinguish between these terms, emotions are short, intense reactions directed at a specific target, while moods last longer, are less intense and are not oriented at anything specific. Gross (2015) explains it as moods being less specific than emotions, highlighting that they more often influence cognition rather than the behavioural response. In a way, emotions then provide insights into our external environment while moods offer insights into our internal states (Saarikallio & Erikkilä, 2007).

2.2.1 Theories behind affective states

When it comes to the conceptual frameworks that form the basis of emotional phenomena and experiences, researchers face difficulties in reaching consensus. Over the last century, there have been two dominant theories in affective studies: basic emotion theory and dimensional theory (Gu et al., 2019). The two theories contradict each other in that they characterize emotions as either discrete entities or as variations along a single dimension. In short, basic emotion theory states that humans possess a limited set of "basic" emotions on a biological and psychological level that, in turn, can be combined to give rise to more complex emotions. While the theory itself is accepted by many researchers, there is no general agreement on the exact number of basic emotions, as some psychologists propose that there are only four primary emotions (fear, anger, joy and sadness) and others have proposed that there are up to seven or eight.

Dimensional theories originally range back centuries, where early studies proposed three dimensions of emotions: pleasant-unpleasant, tension-relaxation and excitation-calm (Gu et al., 2019). Later studies found that the latter dimensions overlap, and since then other theories have been proposed where the dimensions consist of two scales: pleasant-unpleasant and active-passive. One of the most dominant approaches to dimensional theory is one proposed by Russell (1980), termed the circumplex model of affect, where affective states are arranged in a two-dimensional space of pleasure-displeasure and arousal-sleep. In a later study, Posner et al. (2005) reestablish this proposition by asserting that there are two fundamental neurophysiological systems responsible for all affective states, valence (pleasure-displeasure) and arousal (rest-activated). They suggest that all emotions can be explained as a combination between these two dimensions and are

thereby characterized according to their location on the circumplex and which of the four quadrants they belong to (see Figure 2.1).

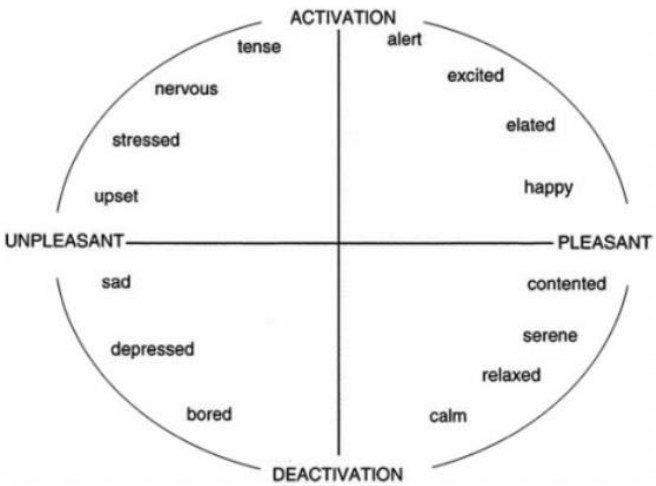

Figure 2.1: The circumplex model of affect (Posner et al., 2015).

The use of Russell's two-dimensional construct is prominent in affect studies (Han et al, 2009). More recent studies have used the circumplex model of affect in a variety of different contexts, including emotion recognition systems (Zhuang et al, 2019), emotion's effect on human-machine interaction (Du et al, 2020), emotion regulation interventions (Dingle et al., 2016) as well as music emotion recognition and classification (Han et al, 2009; Helmholz et al, 2017). The exact names of each of the dimensions vary across different studies, with examples such as valence and activity as well as valence and arousal, but the core essence of each dimension is about the same (Lu, Liu & Zhang, 2006). To circle back to basic emotion theory, Gu et al. (2019) suggest that the circumplex model can accommodate all basic emotions, as they can all be categorized by their fundamental levels of pleasure and arousal.

2.2.2 Affect regulation

The act of self-regulation can be defined as any human effort that is aimed at consciously modifying internal states or responses (Skånland, 2013a). Self-regulation encompasses the concept of affect regulation which, in turn, point to the actions taken by individuals to maintain, enhance or suppress levels of affect, both positive and negative (Baltazar et al, 2019; Saarikallio & Erikkilä, 2007). Park et al. (2019) point out that people employ different techniques for managing and regulating their mood in order to effectively deal with the challenges of everyday life. They argue that what mood-regulation techniques people adopt can either have a positive or negative impact on their mental health, relationships and general well-being. Others argue that affect regulation in itself is essential to a sense of well-being and positive mental health in everyday life (Skånland, 2013a; Goethem & Sloboda, 2011).

2.3 Music for affect regulation

Several studies support the claim that music is a common resource for affect regulation and cultivation, both on conscious and unconscious levels (Baltazar et al., 2019; Skånland, 2013a; Goethem & Sloboda, 2011; Saarikallio & Erikkilä, 2007). Siles et al. (2019) points to this in the sense that music becomes "the engine of how you want to feel". In their study on how users exploit music listening to cultivate affect, findings showed another dimension of affect regulation: the use of music as a tool for responding to moods that derive from specific experiences, rather than solely using it to generate new affective states. Participants reported that music provided a form of communication and enabled them to convey their feelings during particular moments in time, as well as maintain those affective states for an extended period.

Similarly, Park et al. (2019) suggest that music listening differs from other types of media consumption in that it gives insight into what people may want to feel, in addition to what they already are feeling. Siles et al (2019) had similar findings, and mean that music reflects the moods and emotions of a user and vice versa.

A study by Goethem and Sloboda (2011) examined how people use music for affect regulation in everyday life, and found that music is used to regulate a wide range of emotions. Most regulations were aimed at changing negative emotions such as stress, weariness, boredom or sadness to more positive ones, such as happiness, excitement or relaxation. However, some participants also used music to evoke nostalgic or wistful moods. Similarly, Siles et al. (2019) highlight that music becomes a tool for revisiting past moods and emotions associated with particular life events. Saarikallio and Erikkilä (2007) also underline that music is inherently correlated with enjoyment and positive experiences, with a clear association between favorite music and positive emotions.

2.3.1 Individualized music

As highlighted by Skånland (2013b), the impact that music has on a person is not solely determined by the music itself but also by the individual's relationship and interaction with it. Similarly, Howlin and Rooney (2021) emphasize that this relationship between an individual and their taste in music is fundamental to the effectiveness of music-based interventions, meaning that self-selected music is essential when attempting to get an enhanced emotional response.

Several additional research studies have found that individuals' preference and familiarity with music play a significant role in how that music affects their emotional state (Baltazar et al, 2019; Ali & Peynircioğlu, 2009; Dingle et al., 2016). Goethem and Sloboda (2011) underline that people's own choice of music leads to a greater impact on their affective state, compared to when they listen to music that is not of their choosing. For young people in particular, as demonstrated in

Papinczak et al.'s (2015) study on young people's music use, the aspect of self-selected music was found to be an important prerequisite for meeting their music needs. Further, Krause and North (2014) underline that individual control of music listening devices during music listening sessions contributes to more positive moods and consequences than the lack of personal input, suggesting that control of the music is an important contextual factor in music listening.

It is not yet fully understood why self-selected music is more effective than the alternative, especially in the context of pain-management and other music-based interventions (Howlin & Rooney, 2021). However, as Ali and Peynircioğlu (2009) point out, one potential explanation is related to listeners' expectations and how they interpret emotions in music. They underline that increased familiarity to a piece of music and the corresponding potential to anticipate what follows in the music can result in a form of anticipatory arousal.

2.3.2 Context-based music listening

In addition to the individualized aspect of music listening, there are other potential factors to consider when studying emotional responses to music listening. Goethem and Sloboda (2011) argue that when music listening is combined with other activities, the emotional response is not purely musical in nature as the activity itself may also have an impact on the evoked emotion. They mean that music often plays a secondary role in everyday life activities. Similarly, North and Hargreaves (2008, pp.138, 226) suggest that responses to music are context-dependent in that it is always experienced in a specific time and place, with or without the presence of other people, and simultaneous with other activities.

Additionally, North and Hargreaves (2008, p.127) argue that people's emotional responses to a given piece of music can differ greatly from person to person and can also change over time. Howlin and Rooney (2021) recognize this by highlighting that two people can have very different physical and psychological reactions to the same musical piece. The association between tempo and emotion, for example, is not universal as fast music can lead people to experience both an increased or decreased heart rate, and low tempo music can be associated with both relaxation and boredom or sadness.

2.4 Music recommendation

Recommendation in general refers to helping individuals in finding relevant content in a vast array of choices by utilizing the perspectives and opinions of the consumer community they belong to (Helmholz et al., 2017). Effective recommendation results in a minimized amount of required effort and time originally spent on making decisions.

The widespread popularity and evolution of music streaming services and the corresponding availability of audio data has prompted a recent surge of interest in music recommendation and retrieval (Brinker et al., 2012; Martín-Gutiérrez et al., 2020). Bodó and Szilágyu (2018) highlight that in turn, this suggests an essential need for recommender systems that facilitate the search for similar music. This is a central concept in Music Information Retrieval (MIR), where such applications play an important role in providing music streaming platforms with algorithms and services that result in recommendations. (Martín-Gutiérrez et al., 2020).

Agres et al. (2021) underline a specific type of music generation software where focus is laid on interactive improvisations and automatically generated music and interactive improvisation in that the system can learn from recently played tracks and then guide and influence the individual into a specific affective state. However, they also point out that existing music can be used as leverage for music recommendation by generating playlists tailored for the individual.

2.4.1 Context-aware recommendation systems

In order to create a more personalized user experience in regards to music listening, Helmholz et al. (2017) highlight the importance of specifying the situation or context in which the listening takes place and establish the appropriate musical characteristics. They present the idea of context-aware music recommender systems, where focus is laid on recommending musical pieces depending on the user's current situation, context or emotional state. In particular, such systems target contexts that may influence the emotional response of the user and thereby the interpretation of the recommended items. Helmholz et al. underline that such systems should take three main factors into account: musical properties, the context of the user as well as the user's preferences. The majority of existing research on recommendation systems disregard the user's specific circumstances or context and concentrate only on their personal preferences.

2.4.2 Mood-based music classification

Helmholz et al. (2017) propose that music needs to be classified and categorized in order to be used in recommendation systems. One of the most prevalent practices for classifying music is by genre, which has shown limitations in regards to automatic classification due to the blurred distinction between genres. Another more modern approach to classifying music is to categorize it based on mood and emotions, using audio features such as valence, arousal and energy. Similarly, Brinker et al. (2012) emphasize that mood-based music classification can be utilized in order to define musical distance measures, which in turn can define mood-specific song features and mood presets to use in music recommendation systems. This form of computational music analysis, where musical structures are extracted and used for understanding emotional content, has become a significant focus of research in MIR (Agres et al., 2021; Han et al, 2009). Further, Schriewer and Bulaj (2016) describe the intricate relationship between emotions and musical

structure by pointing to musical components and musical anticipation as the main drivers of modulation. They mean that the tempo, rhythm, lyrics and predictability of music are factors that, along with others, can impact listeners' emotional arousal and pleasure and thereby their mood.

2.5 Existing technologies

Throughout history, media technologies have had a significant impact on the regulation and cultivation of affect and mood, as users often turn to technology to amplify or express their moods (Siles et al., 2019). The challenges faced by previous studies on music consumption can in modern times be tackled with the increasing use of both mobile devices and music streaming services on a global scale (Park et al., 2019). IFPI's (2022) yearly report on the role music plays in everyday life around the world states that 74% of internet users listen to music using paid subscription audio streaming services. In addition, the study reports that people on average use more than six different practices for engaging with music, including video, audio and radio streaming.

2.5.1 Young people's use of technology

Kauer et al. (2012) highlight that when targeting young people, technology is deemed a particularly important aspect as they are accustomed to using electronic devices and rely on technology for most of their social interaction. As a demographic, they have been dubbed "early adopters" and are quick to engage with electronic devices in an effortless way. IFPI's (2022) report additionally indicates a higher use of subscription audio streaming services in younger demographics, namely ages 16-24 and 25-34. Young people also use their mobile devices to engage in emotional health optimizing strategies more than any other demographic, including social networking, fitness and listening to music (Rickard et al., 2016). Approximately two thirds of music listening by young people is via mobile devices such as mobile phones. In other words, interventions that involve modern, mobile technology have the ability to attract and engage young people.

2.5.2 Music streaming services

There are currently a variety of music streaming services available: Spotify, Apple Music, Amazon Prime Music, SoundCloud and Google Play Music, among others (Schriewer & Bulaj, 2016). Many of these streaming services offer a diverse set of explicitly mood-related categories that the listener can choose from in order to reach their desired mood. In some cases, they also offer context-based music stations, based on specific songs, artists, genres or moods, that recommend similar songs to the user. These music streaming services are available for listening in various contexts, including in desktop format and on mobile devices such as smartphones and tablets (Hagen, 2015).

2.5.3 Playlists

One of the most fundamental components of any music streaming service is playlists, as they have become leading in the sharing, discovery and gathering of music (Siles et al., 2019). Advancements in technology, and especially regarding mobile devices, has enabled playlists to become an integrated aspect of everyday life (Krause & North, 2014). In addition, playlists offer more control and an increased interactivity for the user, in comparison to listening to traditional albums and their set track order.

A study by Hagen (2015) investigated how users describe and understand their own habits and experiences related to creating, maintaining and utilizing personal playlists. The findings showed that listeners utilize a combination of static and dynamic playlists to organize their music, as well as temporary ones. Playlists that maintain a specific set and order of tracks are static, but can be revived to become dynamic. Dynamic approaches include both updating existing playlists and creating new playlists, where the creation can be as frequent as one playlist per listening session. In this way, music streaming services enabled participants to be able to assemble their long-term or fleeting preferences in a flexible and convenient way. Whether static or dynamic, Hagen underlines the use of playlists to manage the overwhelming amount of music in streaming services and their role in accompanying the listener in daily life.

Siles et al. (2019) argue that playlists can be defined as affective genres, emphasizing that genres are not defined by their substance but by the results of their actions. In their study, they highlight that playlists capture specific moods and emotions that people associate with events and experiences in their own life, making users turn to them to produce, explore and respond to these moods. In this way, they argue that playlists become the ultimate tool to use when cultivating affect with use of music streaming services.

Krause and North (2014) highlight that when creating playlists, listeners often consider specific characteristics such as their mood, track features and the context in which it will be played. In this way, playlist creation can be a means for individuals to regulate their mood. Similarly, Hagen's (2015) study on user's experience with playlists also found that they function as a form of self-expression and are thereby individualized by nature. Additional findings showed that one common practice was to create playlists based on specific contexts, particularly dynamic temporary playlists. In most cases, participants reported to use themselves as the context when curating playlists, meaning that their personal moods and experiences determined the music.

2.5.4 Spotify Audio Features

Research shows that playlists that are designed with the purpose of representing a specific mood or context vary in terms of their Spotify audio features (Duman et al., 2022). Spotify characterizes music tracks using 11 specific audio features: acousticness, danceability, duration, energy, instrumentalness, liveness, loudness, mode, speechiness, tempo and valence (Park et al., 2019). These audio features are meant to give an overview of the composition of the music (Howlin & Rooney, 2021).

Spotify puts their audio features into three different categories: mood, properties and context, where mood incorporates danceability, valence, energy and tempo (Duman et al., 2022;). These four features, along with instrumentalness, have been found to share similarities with typical music chosen for emotional interventions (Howlin & Rooney, 2021). However, Spotify does not disclose the precise meanings and descriptions behind each of the audio features, and the values which are provided by the API could thereby be considered unreliable. Some of the features are not purely audio properties, such as danceability, which rather relates to the use of the music than its features.

Table 2.1: Descriptions provided by Spotify of mood-related audio features[1].

Spotify Audio Feature	**Description**
Energy	Energy is a measure from 0.0 to 1.0 and represents a perceptual measure of intensity and activity. Typically, energetic tracks feel fast, loud, and noisy. For example, death metal has high energy, while a Bach prelude scores low on the scale. Perceptual features contributing to this attribute include dynamic range, perceived loudness, timbre, onset rate, and general entropy.
Instrumentalness	Predicts whether a track contains no vocals. "Ooh" and "aah" sounds are treated as instrumental in this context. Rap or spoken word tracks are clearly "vocal". The closer the instrumentalness value is to 1.0, the greater likelihood the track contains no vocal content. Values above 0.5 are intended to represent instrumental tracks, but confidence is higher as the value approaches 1.0.
Danceability	Danceability describes how suitable a track is for dancing based on a combination of musical elements including tempo, rhythm stability, beat strength, and overall regularity. A value of 0.0 is least danceable and 1.0 is most danceable.
Valence	A measure from 0.0 to 1.0 describing the musical positiveness conveyed by a track. Tracks with high valence sound more positive (e.g. happy, cheerful, euphoric), while tracks with low valence sound more negative (e.g. sad, depressed, angry).
Tempo	The overall estimated tempo of a track in beats per minute (BPM). In musical terminology, tempo is the speed or pace of a given piece and derives directly from the average beat duration.

[1] https://developer.spotify.com/documentation/web-api/reference/get-audio-features

3 Methodology

In this chapter the various methods used to answer the research problem are presented and discussed.

3.1 Scientific Approach

The scientific approach used to answer the research question incorporates a combination of quantitative and qualitative methods. In turn, this implies that both quantitative data and qualitative data was gathered, meaning both numerical data and data in the form of words and descriptions (Sharp, Preece & Rogers, 2019). This can also be described as a mixed methods design (Robson, 2016, p.174). However, the data was mainly analyzed using qualitative analysis with hints of quantitative analysis, meaning that the focus primarily laid on the nature of the data and eventual themes and patterns that emerged.

The primary research method was in the form of a user study, as a prototype was developed and tested on real users for a set time period. In turn, compatible scientific methods were combined in order to achieve data from those users. The qualitative methods consisted of semi-structured interviews, both in the pre-study context and to gather participants' thoughts after finishing the user study. Consecutively, the quantitative data was a result of quantitative methods in the form of indirect observation, specifically interaction logging, in combination with a self-report questionnaire with both open and closed questions.

In this study, multiple data gathering techniques were incorporated in order to investigate the prevailing phenomena from different perspectives. Sharp, Preece and Rogers (2019, pp.260-261, 264, 300) highlight the importance of not focusing on solely one data gathering technique but to use them in combination in order to avoid biases. They also underline the use of triangulation, a concept that involves the use of different sources, researchers, theories or data gathering techniques in an attempt to validate the results using different approaches. The most common form of triangulation is the methodological kind, where the use of more than one method can point to similar findings.

3.2 Method Description

3.2.1 Prototype for user study

In order to investigate the research question, a web-based application was created to act as the prototype for the data collection. A prototype can be defined as a partial realization of a design that allows for exploring and interaction with a certain concept (Sharp, Preece & Rogers, 2019, pp.422, 424, 428-429). Such a manifestation can be seen as an important tool for testing the feasibility of an idea

on actual users. A prototype usually emphasizes a certain characteristic of the concept and plays down others. In this case, the emphasis was laid on the experience of generating a playlist based on certain characteristics. The prototype was of high-fidelity, meaning that it provided enough functionality to closely resemble the final product and thereby offered the opportunity to test the concept in a real context.

The prototype was created to be web-based and thereby able to be run on both mobile and desktop devices such as smartphones and laptops, which made it suitable for the target group. The technologies used to create this application consisted mainly of those included in the MERN[2] stack: MongoDB, Express, React and Node.js. The architecture behind MERN allows for easy construction of three-tier applications where both frontend, backend and database functionality can be built solely on JavaScript.

The main aim of the prototype was to retrieve the user's listening data, allow the user to pick a desired mood and then generate playlists based on that choice. The generated playlists would then be based on the user's listening history, making the playlists customized to the user's music preferences in addition to their desired mood. This was made possible using the Spotify Web API[3], where previous listening data could be retrieved and recommendations based on that data could be generated. The recommendation system as a whole depended on the mood-related audio features of each track, as it served as the basis for finding similar songs and artists with matching mood-related features. Another dependency was on retrieving the listening data of the user, more specifically the top tracks and artists of the last period in order to make the playlists personalized. Spotify allows for a maximum of 25 users per application in development mode, making it possible to perform testing on an appropriate number of participants.

The prototype incorporated Russell's (1980) circumplex model of affect for categorizing each mood and presenting it for the user to pick from. Spotify's own audio features for each track, namely energy and valence, were used to determine the mood of each song. The features of energy and valence were thereby the main connection between the chosen mood of the user and the moods of the tracks they had previously listened to.

3.2.2 Target group

In accordance with the related research presented and the more frequent and use of technologies in the younger generation, the target group of the study consisted of people of ages 18-25. To narrow down the target group further, the focus was laid on individuals that use Spotify as a music streaming service on a daily basis. The choice of Spotify as the specific music streaming service was mainly based on the

[2] https://www.mongodb.com/mern-stack
[3] https://developer.spotify.com/documentation/web-api

availability of its API, which the prototype could be based on, in addition to the fact that it is one of the most commonly used services used among young adults (IFPI, 2022).

3.2.3 Pre-study interviews

Participants were recruited on the basis that they fulfilled the aforementioned requirements and a pre-study was conducted before the user study to confirm this. The purpose of the pre-study was also to establish a baseline for the participants' music preferences and listening habits as well as their view on music and mood. The pre-study interviews were semi-structured, meaning that they contained both closed and open questions, with a flexible structure to the questions, and allowed the interviewer to ask follow-up questions if necessary (Sharp, Preece & Rogers, 2019, pp.269-270). See the questions of the pre-study in Appendix A.

3.2.4 Self-report questionnaire

In order to understand people's real responses to music, several researchers underline the importance of taking the context-dependent nature of music into consideration and study music listening in the everyday situations where they take place (North & Hargreaves, 2008, p.138; Goethem & Sloboda, 2011). In addition, Siles et al. (2019) underline that the act of regulating affect through music listening is constantly repeating throughout the day, every day. Further, Duman et al. (2022) suggest that future experiments that investigate listening habits in compliance with mood levels should make use of Experience Sampling Methodology (ESM).

This technique features individually self-reported data at multiple points throughout everyday life (Rickard et al., 2016). By using ESMs, participants are requested to share their current or recent affective state, thoughts, behaviour and/or situational circumstances at repeated intervals (Fisher & To, 2012). ESMs are particularly useful considering the temporality of moods and the risk of memory error when reporting past situations and behaviours. In other words, participants may have difficulty recalling the details of their music listening episodes, which is where ESMs can be helpful (North & Hargreaves, 2008, p.13).

Self-reported user tests were conducted multiple times throughout the testing period, at regular intervals, by sending out a questionnaire which the participants could fill in. A questionnaire is a commonly used tool for collecting user opinions asynchronously, either analog or digital, without the need for an investigator to be present (Sharp, Preece & Rogers, 2019, pp.260-261, 278, 301). Similar to interviews, questionnaires can have both closed and open-ended questions, but unless carefully crafted, the responses obtained may not result in suitable data. In this case, a questionnaire was used in order to gather their experiences during the given time period and recall the details of their music listening as it was happening.

The quantitative questions of the questionnaire were built on the concept of Likert scales, which are a common use for gauging opinions, attitudes and experiences, particularly regarding the use of a certain product (Sharp, Preece & Rogers, 2019, pp.280-281; Robson, 2016, pp.308-309). The most common way of formatting a Likert scale is to have five fixed alternatives which the participant can choose from, ranging from "strongly agree" to "strongly disagree", where each alternative is then assigned a weight between 1-5. Each questionnaire also had one open-ended question, where the participant could enter any thoughts and experiences that they wanted to share, but this question was not mandatory to answer.

As the questionnaire was meant to be filled out a few times during the testing period, the aim was for the form to be quick to fill out and have a limited number of questions. The questionnaire was created using Google Forms and sent out to the participants by an online link, for a total of three times during the testing period with five day intervals. See the questionnaire in Appendix B.

3.2.5 Indirect observation

During the testing period, observation was carried out in order to collect data regarding the user's activity in terms of creating playlists based on desired mood. In essence, there are two types of observation: direct and indirect, where the first means that activities are observed in real-time and the latter that the activities are recorded to be viewed at a later time (Sharp, Preece & Rogers, 2019, pp.260-261, 301). This study used indirect observation since data was collected automatically without disrupting the user in the testing process. More specifically, the observation took the form of interaction logging, where participants' activities were tracked in a log which could be analyzed after the testing period. The logs included the playlist metadata, including creation datetime and track audio properties, and the data used to generate said playlist in the form of target audio property values and the user's listening history.

3.2.6 Finishing interviews

After the two-week testing period of the user study, semi-structured interviews were conducted on the participants in order to gather their thoughts and experiences on the prototype and its functionalities. Semi-structured interviews imply that the interviewer has a set of questions, either closed or open, to use as guidance in order to cover the same topics with all participants (Sharp, Preece & Rogers, 2019, pp.269-270, 301). However, these interviews also allow the interviewer to use follow-up questions and probe for more information if necessary, making them suitable for exploring different issues with different participants (Robson, 2016, pp.286, 290). The interviews covered four main topics: the participants' usage patterns, the playlists they generated, their relationship to mood and their overall experience of the application. The questions formulated for these interviews can be found in Appendix C.

3.2.7 Data analysis

The last step of the methodology was to analyze the gathered data. As previously mentioned, the data was analyzed using both qualitative and quantitative methods, meaning that the different types of data were processed using different approaches. The numerical data from both the self-report questionnaires and the indirect observation were entered into spreadsheets using Google Sheets, where descriptive statistics were performed as metrics were calculated and graphs were plotted. The non-numerical data collected through interviews were instead processed using thematic analysis, in an attempt to identify and analyze patterns and topics in the data (Sharp, Preece, Rogers, 2019, pp.320-323). This means that the qualitative analysis took an inductive approach, where concepts and themes were extracted and identified from the data.

3.3 Reliability and Validity

When concerned with the limitations of the chosen methodology, there are two fundamental aspects to consider: reliability and validity (Robson, 2016, pp.104-105, 169-171). Validity refers to the extent of which a test measures what you think it measures, and thereby concerns the accuracy of the findings. As previously mentioned, one way to strengthen the validity of the results is to use multiple approaches to gather data in the form of triangulation. In this study, three types of triangulation were implemented: data triangulation, where more than one data collection method is used; theory triangulation, which makes use of multiple theories; and methodological triangulation, where quantitative and qualitative approaches are combined.

In this study, the possible threats to the validity of the findings are mainly concerned with internal validity, which points to the causal relationship between the practice and its findings (Robson, 2016, pp.107, 169). Since listening to music can be deemed a secondary activity to daily activities, the effects of music listening are context-dependent and the results may vary depending on other external factors. In addition, one should consider the subjective nature of music listening as an experience, especially in regards to using general principles. Variations in the data can and should be expected depending on the user and their individual preferences.

In turn, reliability refers to the consistency of the measurements and whether or not they would produce the same results over time or in different settings (Robson, 2016, pp.105, 173). In this case, the limitations concerning reliability mostly regard different types of bias. The study focused on the topic of moods and emotions, which could potentially be a sensitive area for some users. This might lead participants to not respond truthfully or not want to participate in the study, meaning that there is a possibility for response bias or non-response bias. In addition, the use of self-report questionnaires in terms of music listening in everyday life implies that

the data is subjective in its nature and entirely dependent on the participants' self-reports as well as their current circumstance.

Another issue in terms of reliability is the fact that the application is reliant on Spotify's API, meaning that their data is essential for conducting the user study. The reliability of Spotify's data itself, especially concerning the audio features of tracks, also proves to be questionable since the exact meaning and calculations behind them is not clear. Additionally, the historical nature of the data proves to be a limitation. In other words, the use of Spotify's API and the features when generating playlists could lead to inaccurate or unclear results.

For qualitative research in particular, the trustworthiness of findings is a highly-debated topic due to the lack of a standardized way of establishing reliability and validity (Robson, 2016, pp.112, 173). One option is to lay focus on the credibility of the research and thereby on the details on how the evidence is produced. Establishing credibility thereby means keeping a full record of your activities while carrying out the study, including raw data and the details of the data analysis.

One could also point out the generalizability of the findings as a limitation, especially regarding the low number of participants and the subjectivity of their data. In other contexts, situations or times, or using other participants, the results may be specific for only the group studied and not be as generally applicable as desired (Robson, 2016, pp.110, 172).

3.4 Ethical Considerations

When conducting research that involves real individuals, there are certain ethical aspects to consider (Robson, 2016, pp.194, 197). Research ethics refer to a set of principles and standards of behaviors that is founded on the value and respect of participants' rights as well as the commitment to put knowledge first. Since this study was focused on the topic of moods and music, two potentially sensitive areas for participants, actions were taken in order to protect the users' data and privacy. Though their data was collected in logs to be viewed later, any public versions of these logs will not include any information that could be used to identify them, such as names, usernames or emails. Instead, each participant was given an ID which they were referred by.

The Authorization Flow as presented by Spotify, which was used in order to authorize the users and allow them to connect to the music streaming service through the application (see Section 4.1.1), also incorporates so-called Authorization Scopes. These scopes are meant to put a limit to exactly what any Spotify application can access from the user by making it clear in the login-procedure.

4 Implementation

This chapter describes the software that was implemented in order to investigate the research questions. In this case, the implementation consisted of two main parts: the web-based application that acted as the prototype of the study, and the back-end functionality needed in order to log the users' interactions with the prototype.

4.1 Application

This section describes the development of the major components and functionalities of the web-based application, named MOODsic for its combination of music and mood. The software was primarily developed using the MERN stack, which combines both front-end and back-end functionality by use of four technologies: MongoDB, Express, React and Node.js.

4.1.1 Spotify Authorization

In order to base the generated playlists on the users listening history and add these playlists to their library, the application had to connect the users to Spotify and enable them to log in to their account. This was made possible using the Authorization Code Flow[4] provided by Spotify. In essential terms, this means that the user grants permission only once and stays logged in to the application.

The authorization flow consists of three main steps. First, the application sends an authorization request to access the data. Spotify's Accounts Service displays this request as well as what data the application wants to access back to the user and prompts them to login. The user can then either choose to login and authorize access or cancel the request. The two remaining steps of the authorization code flow involve the use of tokens, which are needed in order to make personalized requests to the API.

In other words, the user's only task in the authorization flow is to login to their Spotify account and approve of giving the application access to their data. Once the user clicks the "Connect with Spotify"-button on the start page (see Figure 4.1-4.2), they are directed to the authorization page, which in turn redirects the user to the server where the abovementioned steps are performed. Once logged in, the page reloads and they can carry out the next steps of picking the mood (see Figure 4.3-4.4). If the user wants to log out from Spotify, they can do so by clicking on the button in the navigation bar.

[4] https://developer.spotify.com/documentation/web-api/tutorials/code-flow

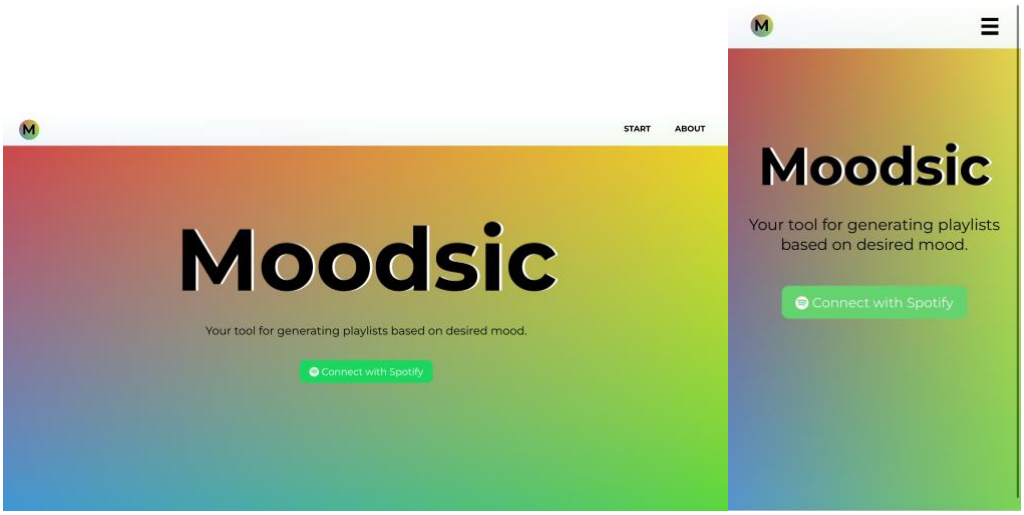

Figure 4.1-4.2: Start page when not logged in, with desktop to the left and mobile to the right.

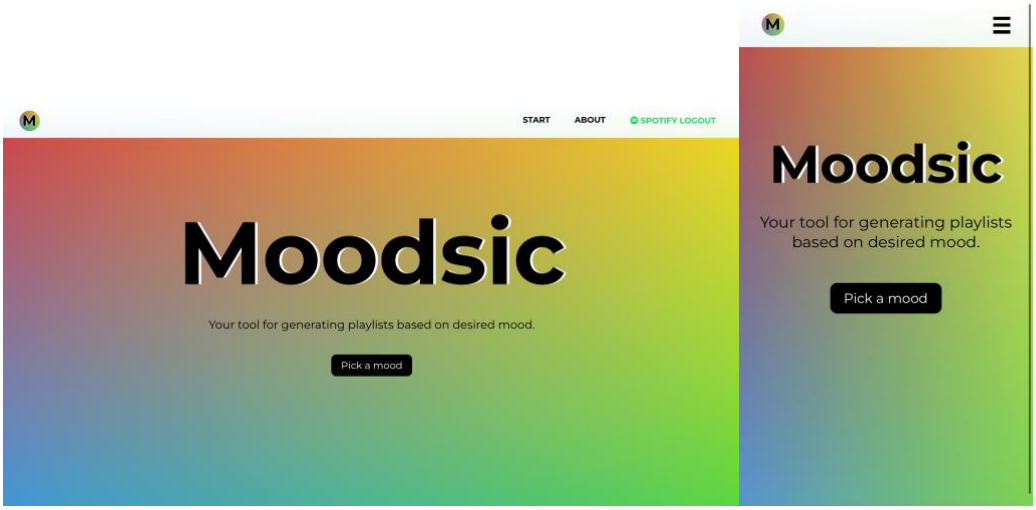

Figure 4.3-4.4: Start page when logged in, with desktop to the left and mobile to the right.

4.1.2 Mood Picker

One of the main intentions of the application was to allow the user to pick a desired mood to base the generated playlist on. A mood picker was thereby implemented, which allowed the user to pick one of 16 moods on a colored graph, see Figure 4.5-4.6. The graph was divided into four quadrants, with 4 moods and one main color for each quadrant. The metadata for each color was gathered in a JSON-document with this graph in mind, and thereby consisted of these 16 moods and their corresponding colors, descriptions and degrees of valence and energy (see Appendix D).

As previously mentioned, each mood was categorized using Russell's (1980) circumplex model of affect, considering its wide-spread use in affective research. Consequently, the mood picker was developed as a translation of this valence-arousal model, and thereby had two axes: energy and valence. The latter was translated into the term "happiness" in order to make it easier for the user to

understand. Originally, each degree of valence and energy corresponds to a number between 0-1. For each user, this was calculated to fit their own levels (see Section 4.1.3).

On each end of each axis a description of the section was implemented, as to visually explain what each mood corresponded to without putting a term to each mood. For example, one mood could be described as “low energy and semi-high happiness”, which would then put it in the green section of the graph. Earlier versions of the mood picker included putting a word to each mood, but as no existing theories were found which supported or could agree on a specific word to each of the 16 mood options mentioned, this was not included in the final version. As the objective of the mood picker was to allow the user to pick a desired mood, a title was also added which asked the question: “How do you want to feel?”. In addition, a description of what the user was supposed to do was added in an attempt to guide the user in the right direction.

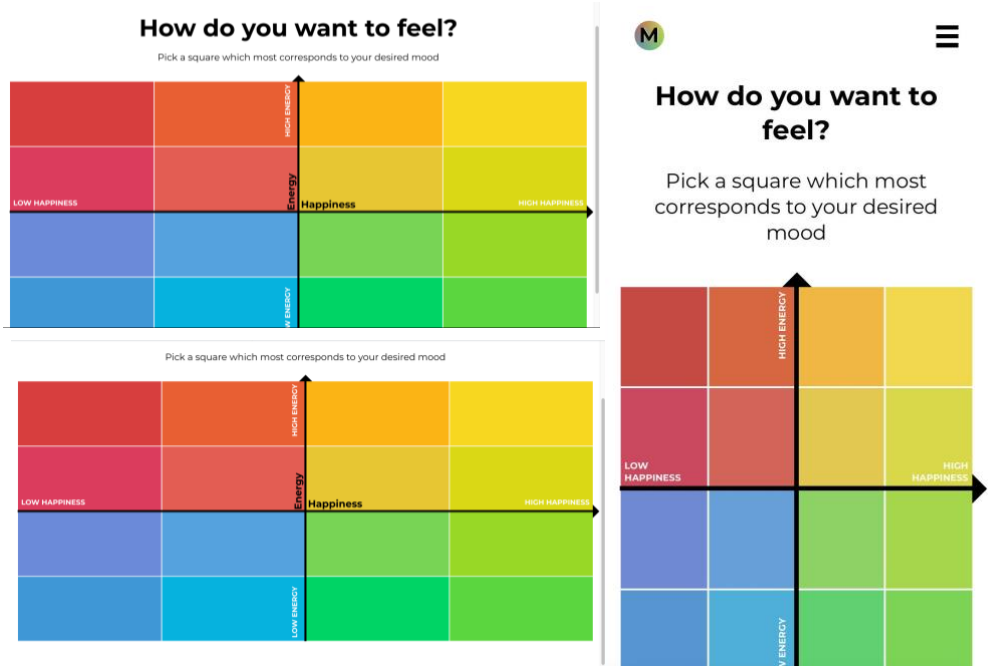

Figure 4.5-4.6: Mood picker page, with desktop to the left and mobile to the right.

Once the user has chosen a desired mood, they are directed to the intermediate page, see Figure 4.7-4.8. The aim of this page was to communicate the choice back to the user in the form of a background with the color of the chosen mood as well as a description of that mood. This page is also meant to give the user the opportunity to go back and change their desired mood before the playlist is generated, as the list is generated and added to their Spotify library once they have decided. The active choice of generating the playlist is then presented to the user in the form of a “Generate playlist”-button.

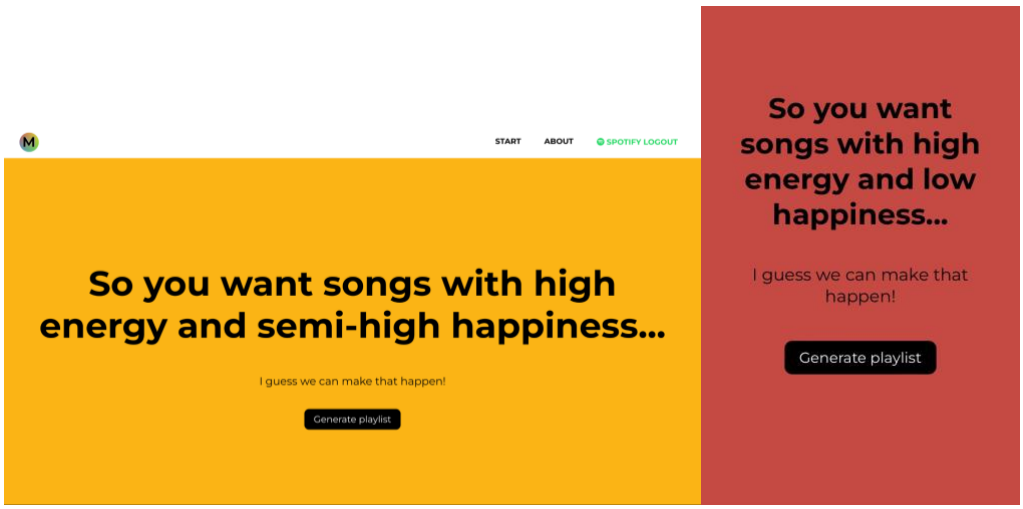

Figure 4.7-4.8: Intermediate page, with desktop to the left and mobile to the right.

4.1.3 Playlist Generator

Once the user has chosen a desired mood and decided to generate a playlist, they are directed to the generator and its loading page. The goal of the playlist generator was to base recommendations on the chosen mood and the users' recent listening history in terms of what they had listened to the most. The first step of the generator was thereby to access the user's 50 top tracks from two periods: the last month and the last six months, which results in 100 individual tracks in total. This was done by accessing the endpoint Get User's Top Items[5]. The duplicate sets were thereafter removed from the set of tracks.

In order to filter these tracks based on the desired mood, the next step was to retrieve the corresponding audio features of each track using the available endpoint[6]. Out of the 11 audio features that Spotify provides for each track, only the five that most corresponded with mood were retrieved: energy, valence, tempo, danceability and instrumentalness (see Section 2.5.4). The two most important out of these audio features were energy and valence, as they were the basis for the mood graph and thereby corresponded to each mood.

In order to make the recommendations individualized, the graph was distorted to fit the range of energy and valence of each user. This was done by finding the minimum and maximum energy and valence levels in the list of top tracks. To make a visualization of this, an example has been made from one of the participants of the study (see Table 4.1). In essence, the distortion meant that each mood was calculated with the new minimum and maximum in mind, making the feature ranges different for each user. These ranges could even differ for the same user, if the calculations were to be based on top tracks from a different time period. In other words, each time a user generates a playlist they are assigned a range of energy and

[5] https://developer.spotify.com/documentation/web-api/reference/get-users-top-artists-and-tracks
[6] https://developer.spotify.com/documentation/web-api/reference/get-several-audio-features

a range of valence which their recommendations are based on. See another visualization of this distortion in Figure 4.9.

Table 4.1: Example of minimum and maximum valence and energy of top tracks.

Feature	**Minimum**	**Maximum**	**Range**
Standard (for both valence and energy)	0	1	[0.2, 0.4, 0.6, 0.8]
Valence	0.078	0.914	[0.2452, 0.4124, 0.5796, 0.7468]
Energy	0.358	0.907	[0.4678, 0.5776, 0.6874, 0.7972]

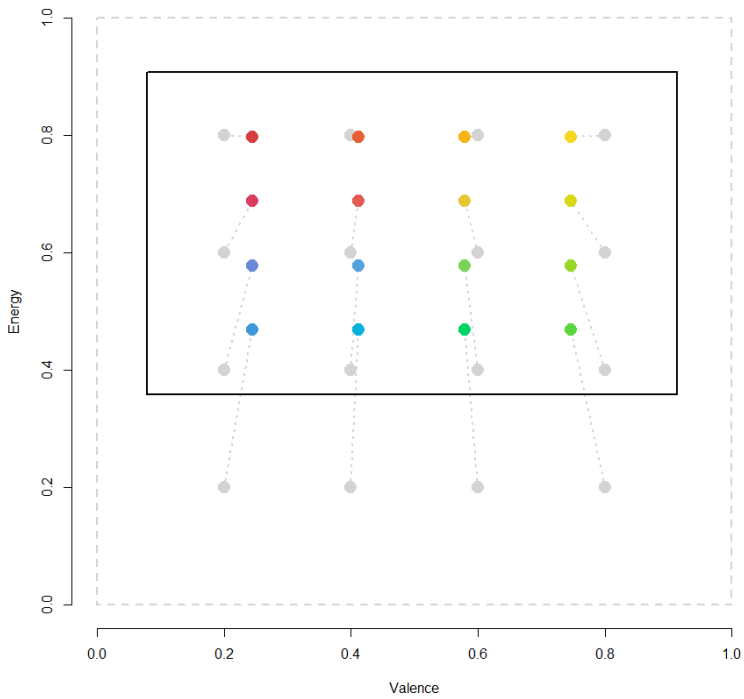

Figure 4.9: Example of distortion of valence and energy.

Based on the chosen mood and its corresponding features, the target level out of these four in each range is then calculated. In this example, if the user chooses the mood where the standard features are [0.6, 0.6], the targets are then [0.5796, 0.6874]. Consequently, these targets are used as the basis for filtering the top tracks based on mood. In order to be categorized to be a certain mood, the track's closest level should be the same as their target level for both features. See a visualization of this example in Figure 4.10.

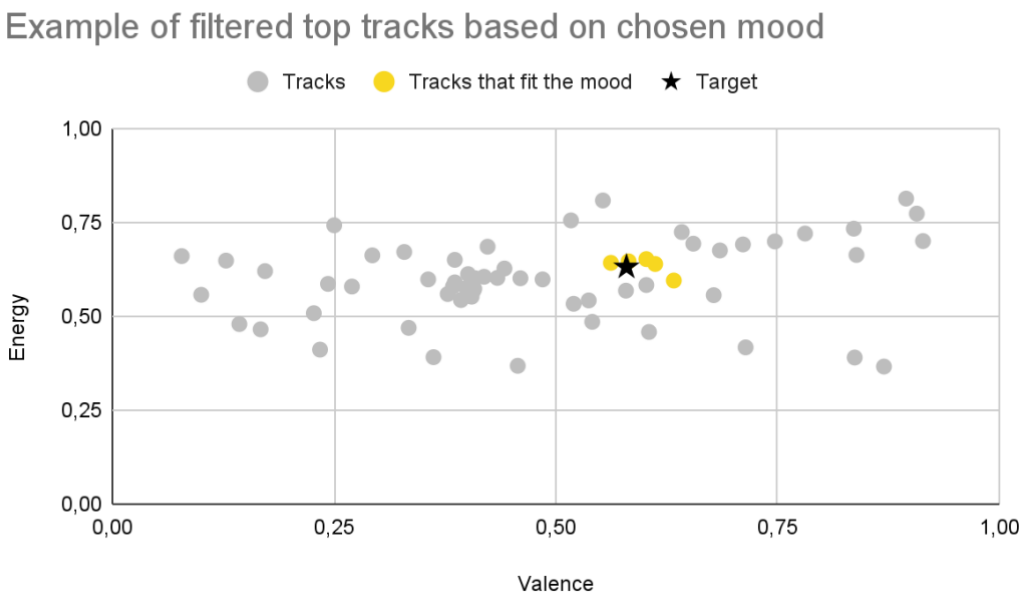

Figure 4.10: Example of filtered tracks based on chosen mood.

The tracks that match the chosen mood then gets added to the list of items to use as basis for the recommendations. If the user has no top tracks that correspond to the chosen mood, i.e. if the target level for energy and valence are not the same as the features of the mood, then the recommendation is based on the user's top artists instead. Should the number of tracks be less than five, the recommendation items consist of a combination of those tracks and the user's top artists. These top artists are retrieved from the API similar to how the top tracks were retrieved earlier, using the corresponding endpoint.

The last step of the playlist generator is to create the actual playlist, which includes a custom playlist name, description and cover. The playlist name includes the current user's first name, the main mood of their playlist as well as the numerical order of the created playlist, such as: "[user]'s happy playlist #3". The description of the playlist points to the current date, what artists it is based on as well as the fact that it is personally generated. The playlist covers were created with use of HTML-canvas, and consists of the MOODsic name and the mood description with the mood color as background. See example in Figure 4.11-4.12.

Figure 4.11-4.12: Examples of cover image.

The recommendation as a whole is then based on four main factors: the user's top tracks filtered on the chosen mood, top artists (if top tracks are too few), target energy and target valence. To get the recommended items, Spotify's endpoint Get Recommendations[7] is used. This endpoint has limitations in terms of the number of recommended tracks as well as the number of items which the recommendation can be based on. This consists of five items in total, in this case using a combination of top tracks and top artists if needed. After accessing the endpoint using the chosen recommendation items, the generator receives a list of tracks which they think fit those criteria in return. The recommended tracks are then added to the newly created playlist in combination with the tracks that the recommendation was based on, meaning that each playlist ends up with a combination of new and previously listened to tracks.

For the user, this process is communicated through a loading page (see Figure 4.13-4.14). Once the generator has finished creating a new playlist, the list is presented to the user by displaying its name, cover image and description (see Figure 4.15-4.16). The user is also presented with two buttons, where one redirects the user to the newly created playlist directly in Spotify and the other gives them the option to pick a new mood to base another playlist on. See Figure 4.17-4.18 for how the generated playlist looks directly in Spotify on both desktop and mobile.

[7] https://developer.spotify.com/documentation/web-api/reference/get-recommendations

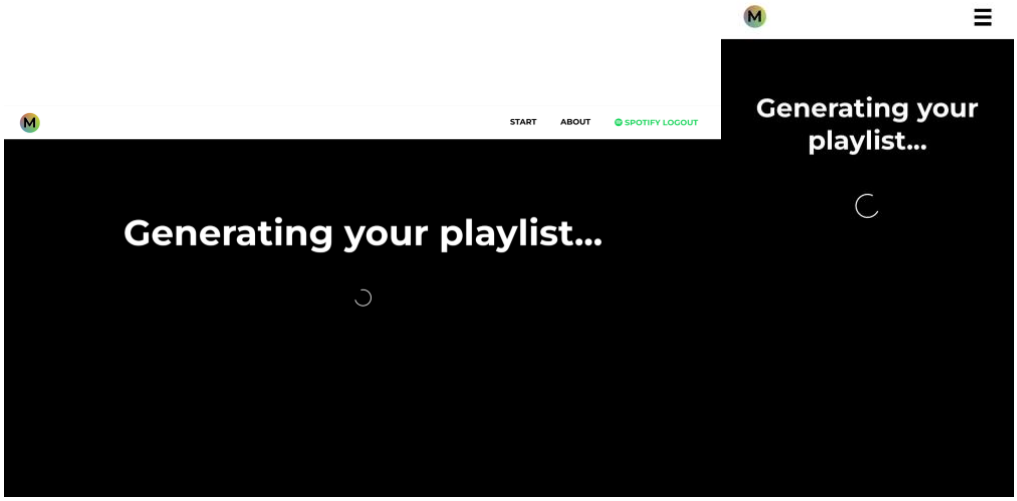

Figure 4.13-4.14: Generator page when loading, with desktop to the left and mobile to the right.

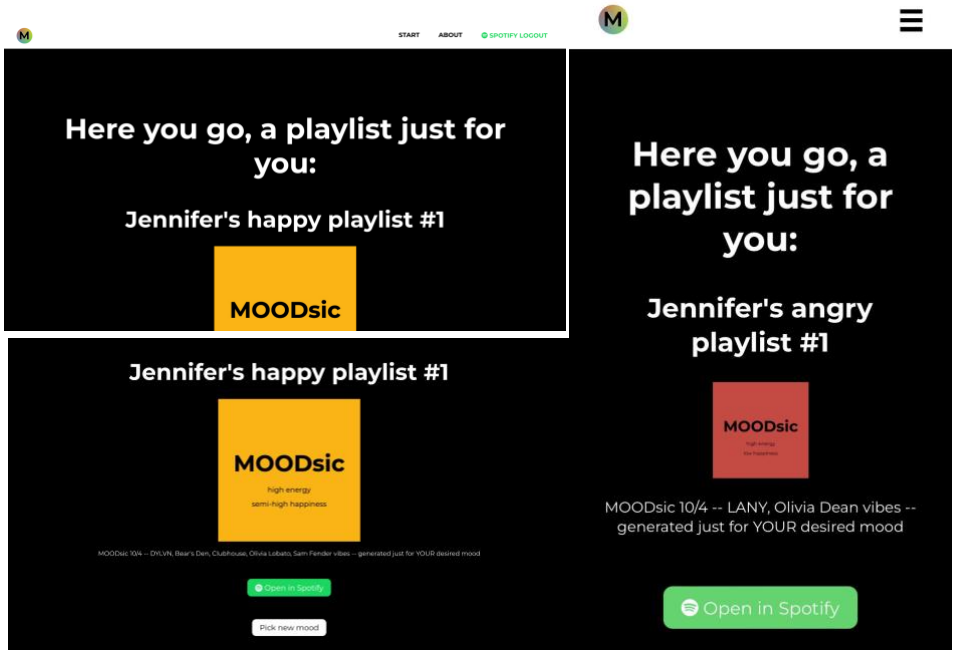

Figure 4.15-4.16: Generator page when finished loading, with desktop to the left and mobile to the right.

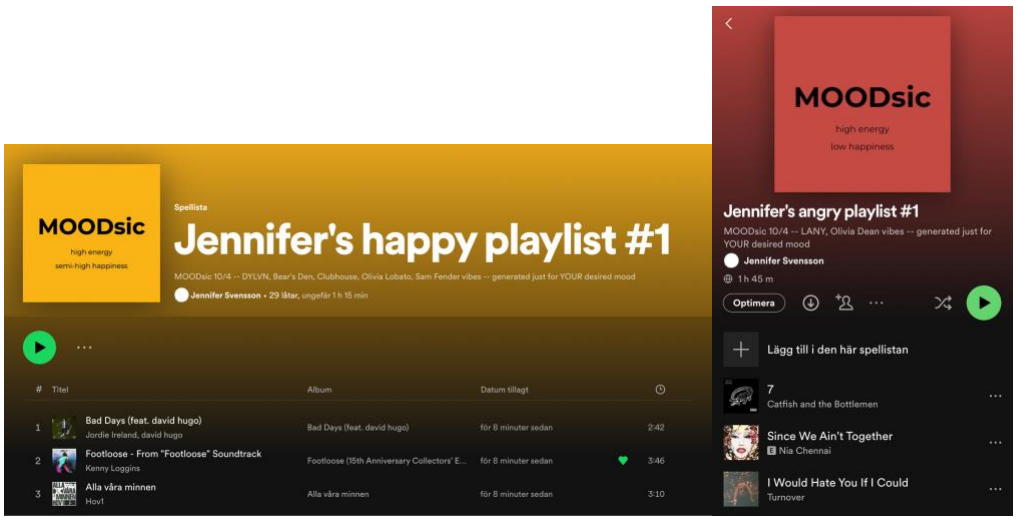

Figure 4.17-4.18: Generated playlist in Spotify, with desktop to the left and mobile to the right.

4.1.4 Design

The colors of the application were mainly based around the colors of each mood, as shown in Appendix D. These 16 colors were divided into 4 main colors: red, yellow, green and blue, which each represented a different main mood (see Figure 4.19). Each mood color was manually selected by making approximated transitions between one color to another, in order to create some form of gradient both between the colors of each quadrant and between the four main moods. The main colors were then used to generate the gradient visible on the start page, to keep the colorful theme throughout all stages of the application (see Figure 4.1-4.4). Aside from these 16 shades, the colors of black, white and Spotify-green[8] were used throughout the application.

#d63e3e	#e85f34	#fbb416	#f7d720
#db3c5d	#e35c53	#e8c632	#dad815
#6a8ad9	#54a2de	#78d454	#98d825
#3e97d6	#04b2dd	#00d465	#5ad63e

Figure 4.19: Colors used in the mood graph.

[8] https://developer.spotify.com/documentation/design#using-our-colors

The font used for the application, Montserrat[9], was obtained from Google Fonts and implemented in all possible stages. The aim was to keep the typeface as clean as possible, to make the experience simple and usable without any distractions from the main goal. The few icons that were used for this project were obtained from React Icons[10], an icon library created specifically for React applications.

Animations were added to all visible components of the application, mostly in order to ease-in the contents of each section once routing to a new page. For the mood picker specifically, animations were added on hover of each box in order to make it clear to the user which mood they would choose. Additional animations were also added to each visible button. However, this hover-animation was not compatible with the use of the application on mobile devices, but the rest of the animations were. This was made possible using the Framer Motion[11] animation library which is meant to be used in combination with React and was thereby fitting for this project.

4.2 Interaction Logging

In order to be able to log each interaction that the user made when using MOODsic, a few things had to be implemented on both client and server side. To log the data and save it in a secure place, a database was established using MongoDB. The application was then connected to this database using the Express server, and endpoints were set up in order to be able to receive and send data to the respective collections.

The most important aspect of the experience that was logged was the creation of the playlist. Each time a participant created a playlist, it was formatted to only include relevant data and then added to the record. The data for each playlist included its ID, name, description, owner, Uniform Resource Identifier (URI), datetime of creation, energy and valence targets, energy and valence levels as well as information about each track of the playlist. For each track, its id, name and foremost artist was gathered along with its mood-related audio features: energy, valence, danceability, instrumentalness and tempo.

In order to get some internal ideas about what the users were listening to, the recent listening records of the participants were also gathered. In practice, this meant making a call to the Spotify API using the endpoint Get Recently Played Tracks[12] in order to get the 50 latest tracks of the user. This call was set to run each time the participants used the application while logged in to their Spotify account. This proved to be a limitation since calls to the endpoint could not be made without some

[9] https://fonts.google.com/specimen/Montserrat
[10] https://react-icons.github.io/react-icons/
[11] https://www.framer.com/motion/
[12] https://developer.spotify.com/documentation/web-api/reference/get-recently-played

sort of user input. As the application was meant to be ubiquitous and not bother the participants more than necessary, the gathering of the listening data could thereby only be carried out each time they used the application. The listening record could therefore be deemed incomplete if a user would not use the application regularly for the testing period.

5 Results and Analysis

This chapter presents the results and analysis of the study, which consisted of a pre-study, a self-reported questionnaire, indirect observation and finishing interviews. The user study, in which the participants were instructed to use the prototype in their everyday life, lasted a total of two weeks. All tests were conducted on the same set of participants (n=8), with 4 male and 4 female. The participants were in the ages 18-25 (M=23.4) and listened to music on Spotify on a daily basis, meeting the criteria for participating in the study.

5.1 Pre-study

Before conducting the user study, a pre-study was carried out in order to investigate participants' existing habits and preferences concerning music listening. The questions formulated for the study can be found in Appendix A. In addition to these, the interviews included questions regarding the demographics of each participant including their name, age and gender. The interviews also included a participation confirmation, where all participants confirmed their willingness to participate in the study and that their Spotify listening data could be recorded for the two weeks of the user study.

5.1.1 Music habits and preferences

The first section of the pre-study interview regarded the participants' music listening habits, see Appendix E.1. The main theme that emerged for all participants was that they listened to music while doing something else, such as walking, cooking, cleaning, studying, working out, driving or taking the bus. All participants reported listening to music at least once every day, which was one of the criteria for being a participant in the study. When asked about the importance of music in their daily routine, all participants deemed it as important to a certain degree. The general consensus was that they could survive without music, but that it made everyday life easier, more fun and less gloomy.

When asked to describe their music taste, almost all participants classified it as broad and varied, but they all had specific genres which they preferred. The answers ranged from pop and rock to house, techno and lo-fi as well as rap, R&B, reggae and funk. What participants preferred mostly varied depending on the situation. Most participants said their music taste was broad, but two categorized it as narrow and one said it's somewhere in between. When working or studying, participants reported to listen to either lo-fi and instrumental music or uptempo tracks and R&B. In their free time, the music they listened to were about the same but ranged to other genres as well, such as rap, hip hop, pop and rock.

5.1.2 Moods and music

The next section of the pre-study interviews concerned the participants' view on the relationship between music and mood, see Appendix E.2. When asked about the importance of music to their overall mood and well-being, the answers varied. The general consensus was that music is an important aspect in changing or enhancing a person's mood. Some pointed out music as being a way to get hyped or focused when working out or studying, and another said it helps to turn a bad day around.

All participants reported having previously experienced a change in mood while listening to music. When choosing what music to listen to when wanting to change their mood, participants used a few different tactics. Some participants had a few different playlists on Spotify to choose from or searched up music that fit their specific mood. Others pointed out that the music they choose varies depending on the mood, but that they tend to try to change their mood for the better.

Participants were asked to give examples of music that they would listen to in order to get into a certain mood, specifically the four chosen moods of this study: Happy, Calm, Sad and Angry. In order to get happy, participants mostly reported wanting to listen to upbeat and festive songs, as well as old, nostalgic favorites and songs you can sing along to or that makes you want to dance. To get into a calm mood, most of the participants pointed to the lo-fi genre of music, as well as instrumental and classical music. As for music when wanting to feel sad, the responses varied a bit more. Some wanted songs where they could relate to the specific type of sadness or the meaning of the lyrics, and others wanted sentimental or typical calm, sad songs. Last but not least, to get into an Angry mood participants wanted genres such as heavy metal, rock and rap, but some pointed out that they don't often want to get mad.

When asked if they currently use or have used any music mobile applications, participants mainly gave Shazam, SoundCloud and YouTube as examples. The majority of participants said that they do not use any mobile applications related to mood, but gave examples such as apps related to training, sleep, study, astrology and entertainment where mood is rather a small part of the experience.

5.1.3 Listening ways

The last section of the pre-study interviews investigated how participants usually listen to music (see Appendix E.3). All participants reported using Spotify as their main listening platform, and some used YouTube and SoundCloud as a complement. In music listening sessions, participants gave varied reports on how they typically interact with their given device, since some said that they put the playlist on shuffle and let it run while others actively choose, queue and skip songs if needed. When listening to music, all participants reported using playlists as their main approach, though some admitted to listening to specific albums and artists as well, and radios to specific songs. The playlists that they listen to were usually

personally created, but some participants also reported listening to Spotify's recommended playlists or ones that were generated by Spotify just for them.

5.2 Self-report questionnaire

A self-report questionnaire was sent out three times during the user study, with about five day intervals. The questionnaire focused on the participants' usage of the application, more specifically if they had generated and listened to any playlists as well as the accuracy of those playlists in comparison to their chosen mood. See a visualization of the answers in Figure 5.1. The results show that the participants self-reported to have generated more playlists in the first round, and less in the next two rounds. However, the participants reported to have listened about the same amount throughout the study period. The accuracy of desired mood in the playlists was high through all rounds, with two outliers as the exception.

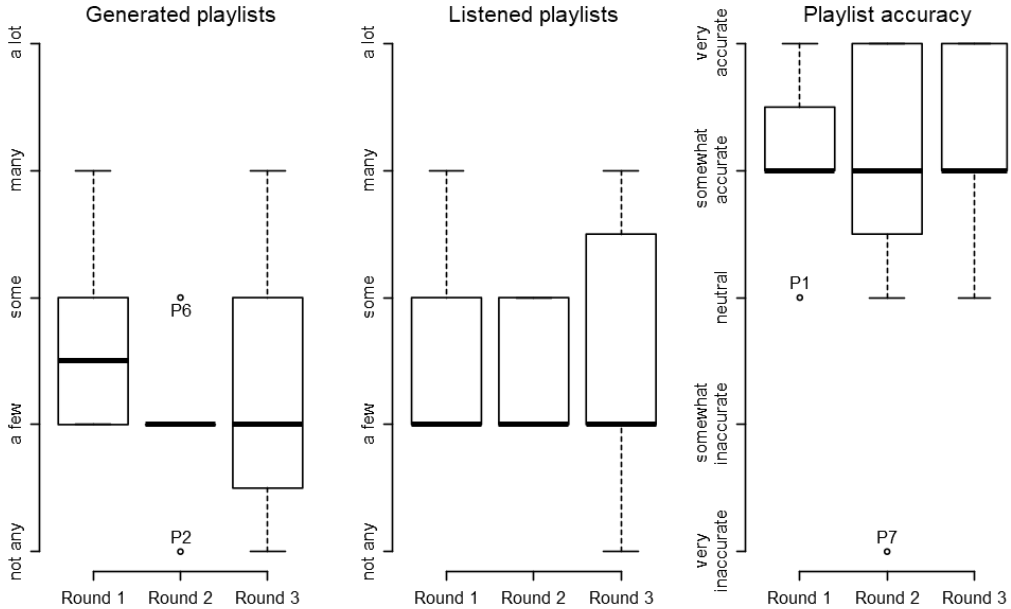

Figure 5.1: Questionnaire results for each question, where the participants responsible for each outlier are identified.

Plots were also created to see how the participants' individual experiences changed throughout the period, see Figure 5.2. Most participants generated and listened an equal amount throughout, however some showed a decrease in their usage, such as participants P2, P7 and P8, while others showed an increase during the period, such as participants P3 and P6. When regarding the accuracy of the playlists, most participants self-reported to about the same level for each of the rounds, except for cases such as P7 who had a dip in terms of all factors. For the third round, P7 filled in that they hadn't listened to any playlists, meaning that they could not answer on the accuracy of the playlists.

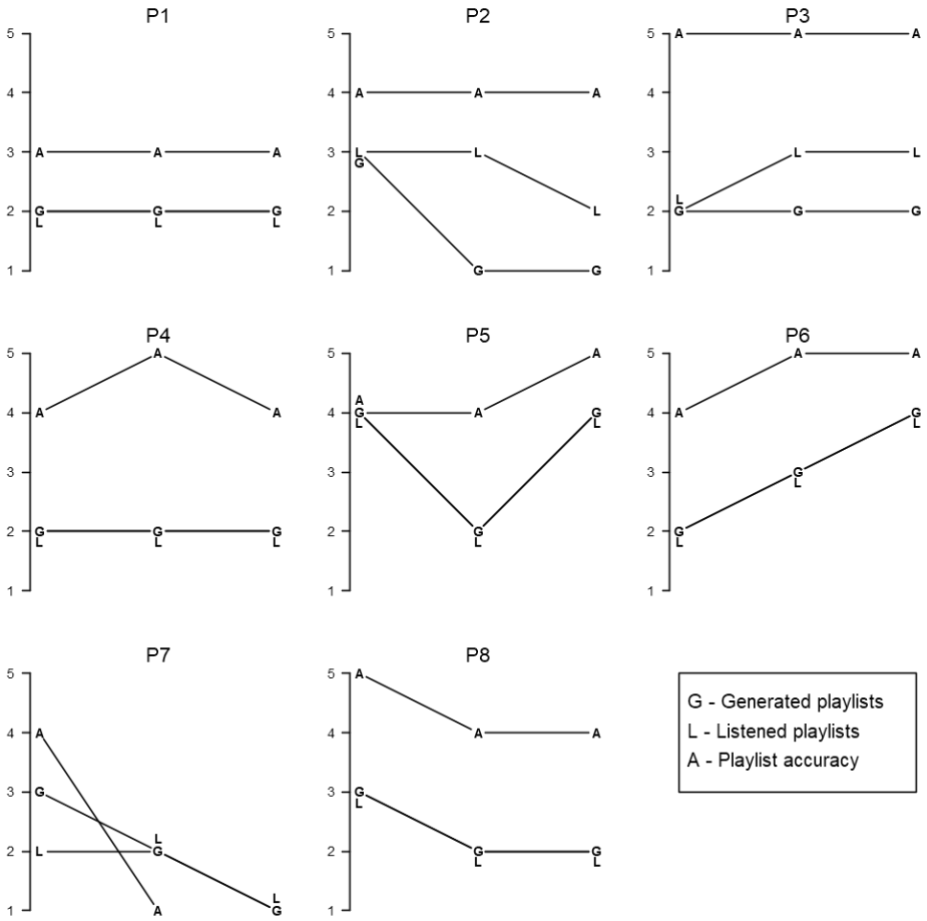

Figure 5.2: Questionnaire results for each participant, over the three rounds.

Additionally, the self-report questionnaire included an open question where the participants were able to elaborate on some aspect of their experience if they wanted to. The answers that emerged elaborated both on positive, neutral and negative aspects of the experience of using the application. See Table 5.1 for a summarized version of the answers and Appendix F for the complete answers.

Table 5.1: Summarized answers to the open question of the questionnaire.

Positive answers	**Neutral answers**	**Negative answers**
"Fun to get unique lists of both new and old songs!" (P2, round 1) "Love it when sleeping and working out" (P3, round 2) "One of the latest creations – calm – had the best selection of songs so far. Appreciate how simple and quick it is to use." (P4, round 2)	"Some songs fit perfectly, others don't belong at all" (P4, round 1) "Add more advanced settings – pick if you want more new or only saved songs" (P4, round 2) "Haven't been listening much lately, just podcasts" (P7, round 3)	"Clumsy to have to use it only on computer" (P1, round 1) "Sometimes playlists forget the chosen mood and throws in liked songs that do not convey the mood at all – happens for about ⅛ of songs" (P5, round 1) "Would have used more if integrated with Spotify" (P1, round 3) "Last generated lists didn't suit my mood, so wasn't excited to make more" (P7, round 3)

5.3 Indirect observation

This section describes the results of the indirect observation, which included data concerning the playlists created by each of the participants and the corresponding features of those lists and their generated tracks. As previously mentioned, the logs also included data on the participants' listening history during the two-week period of the study. However, these logs were deemed to be incomplete due to limitations on how Spotify provides their listening data, and as such they are not included in this section.

5.3.1 Playlists created per mood

Overall, during the two-week testing period, the participants created a total of 47 playlists. All but two of the 16 possible moods were explored into playlists, and the majority of the playlists were either of the happy or calm type (see Figure 5.3). The two moods most chosen were high energy and high happiness as well as low energy and high happiness, with 7 and 6 playlists, respectively. The overall mood that was least picked was angry, with only 7 playlists in total.

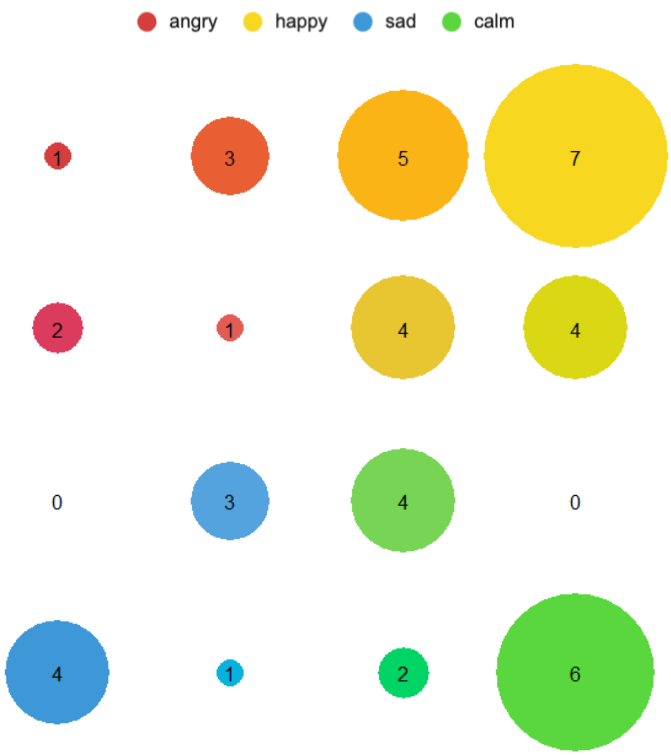

Figure 5.3: Plot of number of created playlists per available mood (n=47).

5.3.2 Playlists created over time

The participants created the most amount of playlists in the beginning of the testing period, with at most 12 playlists in the first day (see Figure 5.4). During the rest of the period, the creation of playlists varied each day, with some days having no created playlists.

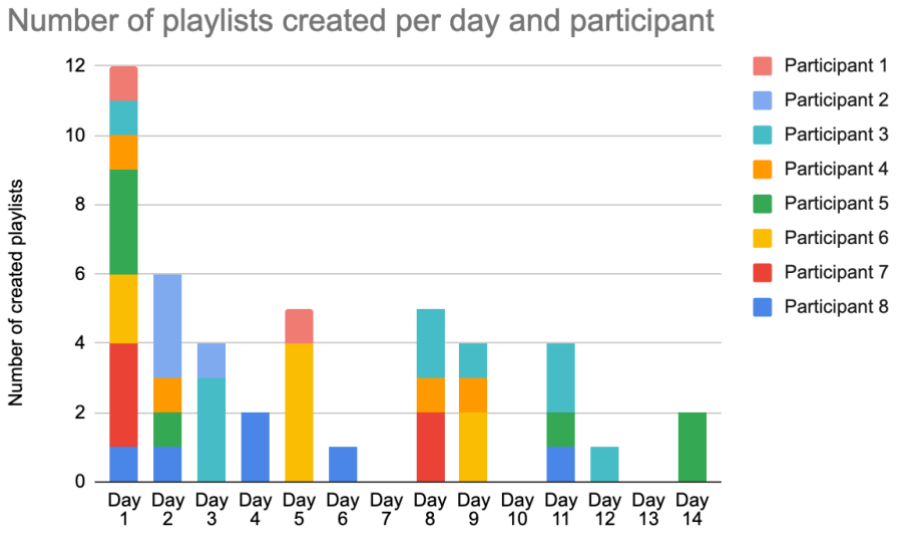

Figure 5.4: Plot of playlists created over time.

5.3.3 Playlists per participant

In an attempt to find patterns in the playlist data, each of the participants' playlists and their respective tracks were plotted along a valence-arousal plot, along with the corresponding target for each playlist (see Figure 5.5-5.12). Note that the grid for each plot is distorted to fit the participants' calculated ranges of valence and energy.

Analysis of these plots enables them to be divided into a few different themes. One such theme is the overlaps of participants choosing the same mood and getting about the same results for multiple playlists. This was mostly common for P1, P3, P6 and P8. For other participants, such as P5 and P7, almost all playlists had a wider distance range between tracks. The rest of the participants had a mix of playlists with spread and compact tracks in terms of their features of valence and energy. Another theme that is made visible in these plots is that some moods were more scattered and spread than others, specifically those with playlists corresponding to a calm mood (green in these plots). However, some of the participants got similar results for playlists made with a happy mood in mind (yellow in the plots).

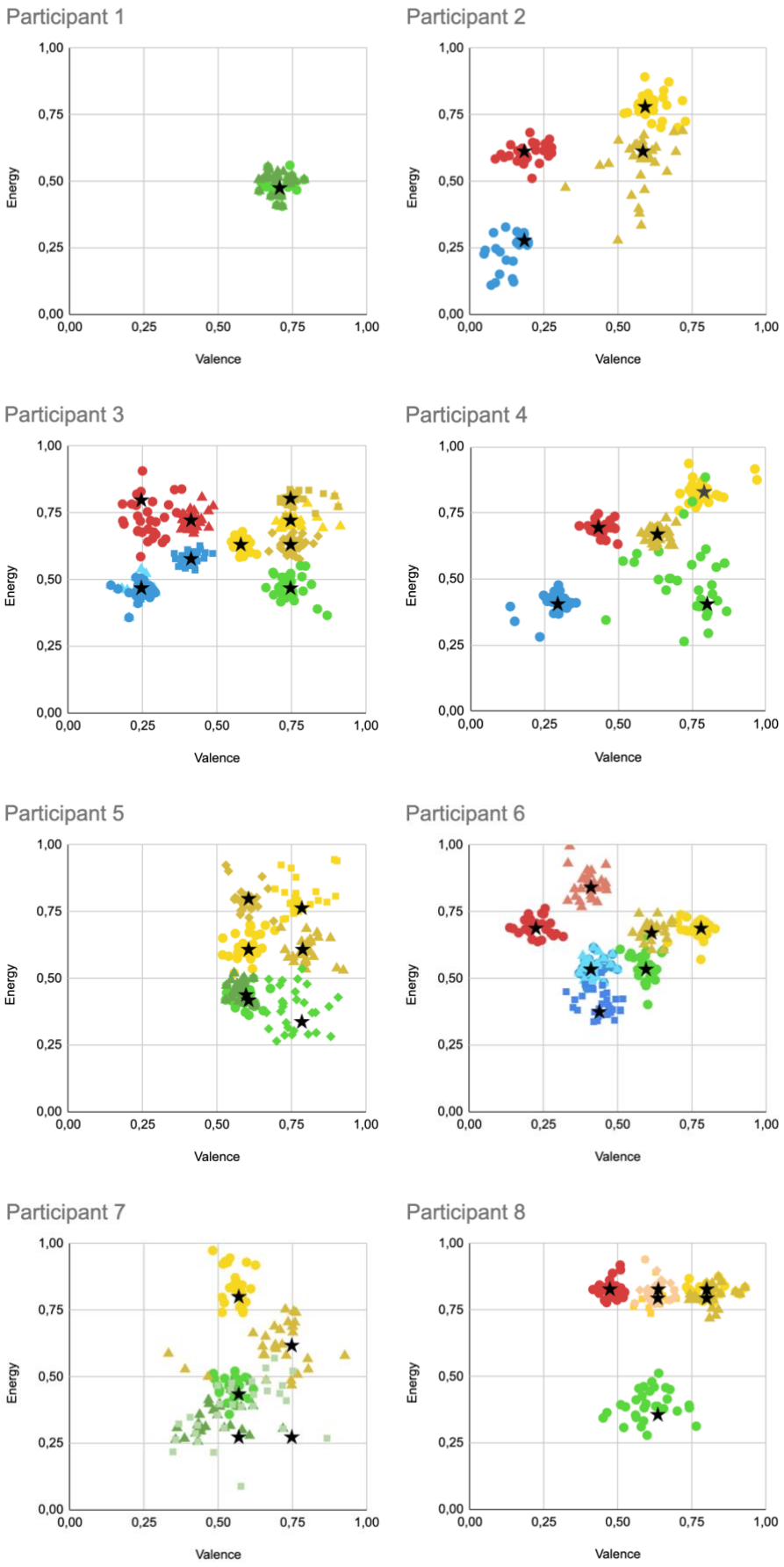

Figure 5.5-5.12: Generated playlists with tracks and targets, per participant.

Additional independent visualizations were created for all of the playlist for each of the participants, see Appendix G. In these graphs, the distorted ranges at the time of the playlist generation are plotted in relation to the target, while also displaying the outline of the tracks in each playlist. In addition, the error of each playlist was calculated, meaning the mean Euclidean distance of each track from the target. The errors that are highlighted in red are those over 0.9 and underlines the more spread cases. Analysis of these plots further indicate that P5 and P7 generated more spread playlists and errors than the other participants.

5.3.4 Difference between target and mean

The difference between target and mean of both energy and valence for all playlists were visualized with the use of boxplots, see Figure 5.13. These plots show that the median difference between target and mean is less than 0.01 for both valence and energy, with a mean of 0.02 for both features. However, these plots also visualize the outliers in terms of what playlists were generated with a higher average difference between target and mean.

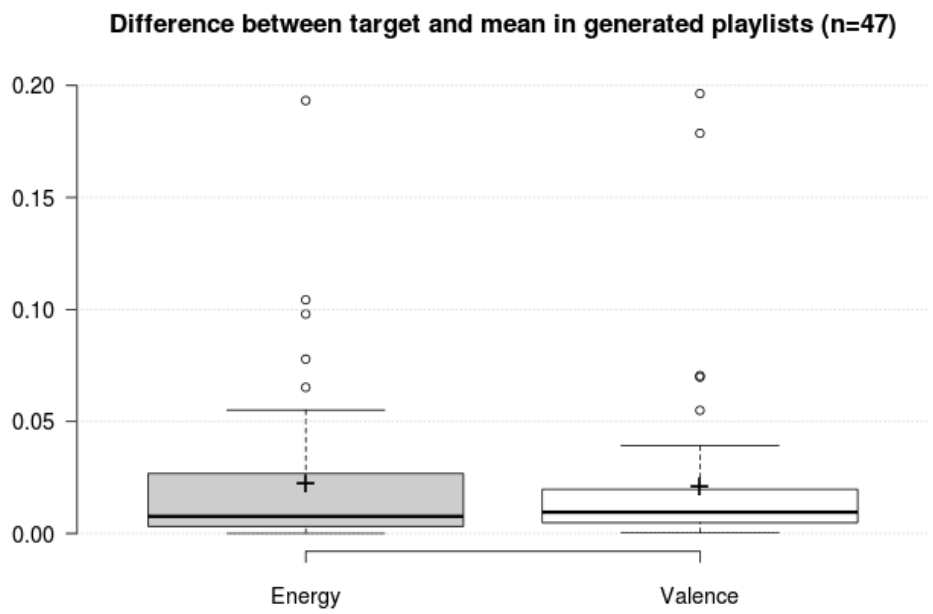

Figure 5.13: Boxplot of difference between target and mean energy and valence in playlists.

5.4 Finishing interviews

The fourth and final data gathering technique was in the form of semi-structured interviews where each participant was questioned on their overall experience of the prototype. In particular, the interviews focused on the users listening patterns, the playlists that they generated both in regards to content and in comparison to the desired mood as well as their experience of the application interface and functionalities. All questions can be found in Appendix C, and all complete answers can be found in Appendix H.

5.4.1 Listening patterns

The first part of the finishing interviews concerned the patterns in which the participants used the application, where the majority said that they used it every few days or twice a week, on average (see Table 5.2). Two participants said that they used it more in the beginning of the period (cf. Figure 5.4), and one participant admitted to not using it that often. Overall, the participants reported to have listened more to the playlists than to making new ones.

Table 5.2: Summarized answers about usage of application.

	How often did you use the application?	**How often did you make playlists?**	**How often did you listen to playlists?**
P1	Not that often.	One, two a week.	Listened to every playlist 2 times.
P2	Not very much. More in the beginning.	5-6 times, in the beginning. then listened.	In the beginning every or every other day, in the end not at all.
P3	Once or twice a week, maybe more.	Every time I needed them.	Every time I created them.
P4	Every few days.	Every few days.	Right after creating them.
P5	Every two-three days.	Every third day.	Once after creating, then once two days later.
P6	Pretty often. Not new ones every day, but used the ones I had quite a bit.	Every second/third day. When I felt I was in a different mood and wanted to.	Pretty often, like every other day.
P7	More the first week, but otherwise once a day when I thought of it.	Every other day.	Listened right after creating then made a new one next time.
P8	Twice a week.	Every time I used it.	Listened right after creating. Listened to each one once, except for one Happy.

When asked about their typical usage patterns, the answers varied from using the application while studying or working, to using it when working out at the gym, to using it while doing household chores such as cooking (see Table 5.3-5.4). Three participants said that they mostly used it in the evening, one in addition to using it in the morning, and another participant said they only used it during the day. As for the overall impact that using the application had on the participants' existing listening habits, the main theme that emerged was that they listened to more new, unknown songs and artists instead of only familiar songs. In addition, some

participants reported rediscovering old songs that they had forgotten about. However, most participants did not report to listen to more music than usual, just explored new artists. When regarding which playlists they listened to the most, the generated ones or their existing personal playlists, the reports were varied as half of the participants listened more to the generated ones, one participant listened more to their existing personal playlists, and the remaining three listened about an equal amount to each type.

Table 5.3: Summarized answers about usage impacts.

	Typical usage patterns	**Impact on listening habits**	**More music? Exploring more?**	**Listened more to generated or own playlists**
P1	When walking, working out	Rediscovered songs, but many songs you recognized	Not more than usual	Listened more to these than own, but it's the same songs
P2	Studying, at school	Not just listening to own playlists, found new songs that I liked	Not more than usual	More on these in the beginning, then switched to own
P3	Based on what I was doing — Studying, sleeping, exercising, reading	Don't listen constantly, so habits haven't changed	No, didn't really change	More to the generated
P4	During the day, at work	Listened to more unknown songs, got reminded of certain songs	No, about the same as usual	More on own
P5	When doing something else – In the evening, when cooking, doing dishes, driving	Listened to more different music	Not more than usual	Both, equal amount
P6	When getting ready in the morning, cleaning and cooking in the evening	Found new good songs, more fun to listen to	Not more than usual	More to the generated
P7	In the evening, when cooking or studying, wanting a certain mood	Found new artists to listen to	Listened to new, different artists, explored moods more	50/50
P8	When working out, cooking, walking	Found new artists	About the same amount	More on these

5.4.2 Generated playlists

The next section of the finishing interviews explored the accuracy of the generated playlists in comparison to the participants' musical preferences and expectations of genres and artists (see Table 5.4). All participants found that the playlists were accurate to their music taste, mostly in the sense that they included songs that they recognized and would listen to otherwise. However, they also reported playlists which were particularly accurate or inaccurate in comparison to others, but what mood those playlists conveyed were different for each participant, ranging from calm and sad to happy, upbeat playlists.

Table 5.4: Summarized answers about usage impacts.

	Playlists reflective of music taste	**Particularly accurate or inaccurate playlists**
P1	Yes, playlists had songs I would listen to and recognized	Wouldn't call the happy songs happy – pretty violent, but guess that's the happiest of my music taste
P2	Yes, accurate for my music taste. But Sad list just become lo-fi, I listen to other music too but guess that's what I listen to the most	When I wanted calm, I got it – very accurate. But sad is also sad lyrics, not just the song. Happy was mixed, high tempo. All but Angry were kind of accurate.
P3	Yes, very accurate	Didn't think about it, don't have a good grasp of own songs. But it was a good mix.
P4	Majority of the songs did, that you had already saved. Other stuck out, where the lyrics and the message of the songs were not taken into consideration	Happy playlist suited the mood, it was mixed with more hyped songs. Songs that you would never put in the same playlist yourself.
P5	Yes, reflected my music taste. Sometimes not the mood.	Yes, felt like 50/50, either on point or it didn't known how to filter those songs
P6	Yes, even had songs I hadn't listened to in years	The Sad lists were probably the best
P7	Songs were good, but it was a little off on whether they matched the mood. Different on low or high	Chose Happy and got a depressing list, so didn't fit very well
P8	Matched the uptempo I wanted, when at the gym. All matched my music, except for one with a lot of Norwegian songs	Happy, high energy was most similar to my music taste and expectations

When asked about certain elements that were deemed missing from the playlists, the answers varied (see Table 5.5). Two of the participants found certain playlists to include only one type of music, such as lo-fi for sad and lack of upbeat music for happy, and thereby felt that the playlists were missing what they expected the mood to be. One participant felt that their most listened to artists were missing from their playlists. The remaining five of the participants could not think of any missing aspects, with one highlighting that it's due to the fact that the playlists were so mixed.

In contrast, when asked about what genres or artists to exclude from the playlists, most of the participants could not name anything specific. Two of the participants mentioned that they wanted more variety in their lists, while another said that they would like to exclude the recommended songs. Two other participants pointed to some specific aspects that they thought did not fit the mood and thereby wanted to exclude those. In regards to the importance of having a wide variety of artists and genres in their playlists, three of the participants said it was very important, two said it was pretty important and the remaining three that it was not that important.

Table 5.5: Summarized answers about aspect to include and exclude + importance of variety.

	Missing genres, artists, songs?	**Genres or artists to exclude?**	**Importance of wide variety in playlists**
P1	No, probably some old artists that you can't think of. Overall fits well	No. Maybe could have been more sorting based on genre	Pretty important. Likes many different artists, boring if just one album in playlist
P2	Yes, on my Sad list is was only lo-fi – would have liked songs about sad stories. In Angry, artists were not angry	Not that I can think of.	Not that important, good music is good music. Can listen to only one artist with a lot of good songs.
P3	No, not from my perspective	Genres and songs were accurate to my mixed music taste, so difficult to know what to exclude	Not really important. Don't really explore other genres, have my own style and stick with it
P4	It's so mixed up, you don't really think about what's missing	Nice to have more familiar songs. In theory good with recommendations, but you skip them	In one list it should be same vibe, gives me the same feeling
P5	No, not that I know. Might have wanted other songs from the artists that fit in	The depressing songs. Brushed past them because they didn't fit	Pretty important, every list I have is a mix of everything, not just one artist
P6	No, can't think of anything	No, not that I can think of. It was good music	Very important, nice to get a variety and not get bored
P7	Many lists that weren't that upbeat, would like more cheerful lists	Possibly some, nothing specific. Maybe more variety in artists	Very important
P8	Yes, listens to Twenty One Pilots quite a bit but they were missing. Was mostly new artists, not many I listen to	Norwegian music, otherwise no	Very important, likes to find new artists

Since the generated playlists included both new and familiar tracks, the interviews included questions about the familiarity of these tracks and how appropriate the inclusion of these songs were found to be (see Table 5.6). The theme that emerged here was that the participants recognized most of the songs and the artists in the playlists, although they had not actively listened to all of the familiar tracks. A few of the participants reported that their playlists included more new content than familiar, but in these cases the recommended songs still matched the style of the others. Most of the participants thought that the inclusion of the new songs was appropriate and that they fit in well, although some participants pointed to cases where some songs didn't match the desired mood or genre. Another participant revealed that they judged the new tracks even before listening and compared them to the ones they recognized, almost not giving the tracks the opportunity to fit in.

Table 5.6: Summarized answers about familiarity and appropriateness of playlists content.

	How familiar was the playlist content?	**How appropriate was inclusion of new songs?**
P1	Recognized most of them. Around 20% new on each list, but matched the others	They were appropriate, same vibe
P2	On a scale of 1-5, 4. Recognized majority. Some I listened to, some from the top chart, and some similar artists	Fit well in Sad, but not so well in Angry. Maybe I don't listen to angry music
P3	A vide variety, a mix. Recognized some, but also songs that were completely new and fit in	Very fitting.
P4	Recognized most of the artists, though I don't listen to all of them. Many songs felt like they were from my library.	The ones you knew were appropriate, the others didn't fit. You compared them and judged the new, didn't give them a listening chance
P5	Recognized 60-80%. More that I knew, but some new. Familiar in that it's similar to what I've listener to too	Appropriate, they fit in, didn't stand out. Except for the ones that didn't match the genre I thought matched the mood
P6	Most of them I had heard before, or years ago, but still many new songs	Thought they were great, it was fun to get something new
P7	A lot new, only a few songs I had heard before. But based on known songs, so it matched the style. Felt familiar, but was still new.	Yes, in general they were very well generated.
P8	Fairly familiar. Not many that you recognized, a few in each list. Pretty much new.	Very. They fit in with what I was doing and the mood I wanted to do.

When asked if they discovered any new music through the generated playlists, six of the participants affirmed that they had found something new (see Table 5.7). One of the participants said that they usually don't take notice of new music since they don't use their own lists. The remaining participant reported that rather than discover new songs, they rediscovered songs that they hadn't heard in a while. They underlined that when you get an old song you've listened a lot to before and forgotten about, you get excited and listen on repeat again. All participants found these new discoveries to be a positive experience, using words such as nice, fun or cool, and liked that they could find new music and broaden their music taste without having to actively search for artists themselves. The participants all reported to have added the newly discovered songs to their own personal library, except for the participants who already had their rediscovered songs saved and those who didn't use their own playlists.

Table 5.7: Summarized answers about new discoveries.

	New discoveries?	**Thoughts about new discoveries?**	**Added the new songs to your personal library?**
P1	Yes.	Always nice, I like to broaden my music taste	Yes, a few songs
P2	Yes.	It was fun. I was happy when I found something new to add to my personal playlist	A couple, can't remember exact number
P3	Yes.	Good, fun. Don't like looking for new things, just want them there. Nice to not have to search and find it yourself	No, I half-ignored my own playlists and only listened to the ones I made. But if I would do it, yes.
P4	Not discover, but rediscover songs I hadn't listened to in a while	No matter the mood, If you get a song you've listened to before, you feel like "oooh" and grind them again	The ones I knew I had already saved somewhere, in album or playlist
P5	Yes. Two.	It felt good.	Yes.
P6	Kind of don't have my own list so I don't notice that, but it was good songs	Very good. You get more variety. Fun with new things when you tire of what you listen to	No, I never use my own lists. I listen more to the radio function.
P7	Yes.	Cool. It's nice to not have to actively find artists elsewhere. You broaden your horizons a little.	Yes.
P8	A lot of new songs by artists I had heard just one song before. Incredibly fun, then I get other songs.	Fun.	Yes. Added one, a new release that I didn't know about. And a french song from my calm list.

5.4.3 Playlists vs. chosen mood

The following section focuses on the generated playlists in comparison to the desired mood that the participants had chosen (see Table 5.8). Participants presented mixed opinions on how reflective the generated playlists were of the desired mood. About half of the participants thought that it was pretty or fairly accurate, two thought it was very varied depending on the mood and one thought it was always very fitting for the mood. For some, certain moods matched better depending on their music taste. Two participants argued that their own musical preferences might be the reason why the moods did not match, claiming that they only listen to depressing or violent music.

All but three participants could name specific instances where a generated playlist did not match their desired mood, where most participants pointed to their angry playlists. Two participants underlined that there were only specific tracks or parts of the playlists that did not match. In such cases where the playlists were deemed inaccurate, some of the participants said that they simply made a new playlist and got a better result, or switched to a previously generated playlist that they knew

were good. Two participants listened to the given playlist anyway, claiming that it was still good music, or listened only to the songs they knew were good.

When asked if they ever experienced a change in mood while or after listening to a generated playlist, the answers varied. Two participants said that they weren't affected in that way. Three others reported that they got into a better or calmer mood after listening to the happy and calm playlists, though one participant pointed out that the change in mood might have been affected by their external circumstances as they mostly used it while working out. The remaining three participants said that they rather used the tool to maintain or amplify their current mood rather than change their mood completely. In addition, one participant reported that they got calm even though they chose a sad playlist, thereby changing their mood to something other than what they originally chose.

Three participants pointed out that the message of the songs should be taken into consideration when choosing tracks based on mood, highlighting that even though a track is musically fitting for the mood the lyrics do not necessarily match. However, they all gave different examples of moods where this should be considered. Although, one participant pointed out that they feel the opposite about this in terms of happy songs, where the important aspect is not the lyrics but rather that it creates a good feeling. For Angry, that participant wanted it to take the genre of the music into consideration instead of what they deemed to be only the tempo.

Table 5.8: Summarized answers about playlists and mood.

	Overall reflective of desired mood	**Instances of playlists that did not reflect desired mood**	**Thoughts about that playlist**	**Experience a change in mood while or after listening?**
P1	Fairly good. Hard with the music genre I listen to. The plot of the happy songs were not that happy. Otherwise it fits.	No, not that I can think of.	-	No. I'm not that easily affected, not much can change my attitude.
P2	"Happy" made me happy. "Angry" didn't make me that angry, and "Sad" made me calmer.	Angry wasn't quite as I thought, then I think like Death Metal. But Happy was happy, that fit the best.	No big deal. I listened to it anyway, it was still good songs.	Sad calmed me down, so just wasn't the right mood. Should check the lyrics, if it is about something sad. Think listening to the happy playlist made me happy.
P3	Varied a bit. Sometimes very accurate, sometimes a song didn't fit. But only a few that stood out.	No. Not a full playlist.	-	Fell asleep once. At the gym I got hyped or focused on what I was doing like I wanted. Not changed mood, but got into the mood I wanted.
P4	Pretty well when the message of the song matched the mood of the playlist. For calm, you have to think of the message of lyrics even though it is musically calm.	Not whole playlists, just parts. Sad had songs with pep vibe, but slower. For Happy, the important thing is that it gives a good feeling. For Angry, consider genre instead of the tempo.	Didn't stay too long, focused on the songs you knew. And go into their song radios and find and queue similar ones that you knew.	No. Wouldn't say that.
P5	Pretty accurate, except for songs that reflected the opposite. Might also be that I listen to depressing songs when I'm happy.	Yes, once got a bland playlist that didn't reflect what I chose at all. Somewhere between happy and calm, but still got angry music.	Bit of a shame, but it was just a matter of making a new playlist and getting a better one.	Yes. Most of the time I got better - mostly made happy and calm playlists and got in that mood, it became more fun to do stuff.
P6	Very fitting. There were always songs I liked and they fit the mood I chose.	Angry had Swedish songs that didn't fit in. One Angry was great and so accurate, not the other	Then I changed to the "Angry #1", because I knew it was right.	Don't change my mood that much. Most of the time you take a mood and keep that mood.
P7	Some moods matched better. Sometimes very good, but some lists did not match at all. But I could just make a new list then.	Yes, I made an Angry playlist that was weird - not bad - but didn't match what I would say is angry music. A calm list was very depressing.	Listened to the first playlist anyway - many new songs, not the wanted mood. Bit off to listen to, I knew what I wanted and didn't match.	Yes. Mostly made lists that I thought matched my current mood, so lists that reinforced my current mood instead of completely changing it.
P8	Mostly chose high energy and happy and all had that tone. Angry had angry text, calm was mostly no text. They matched that mood pretty well and what I expected when I chose that mood.	No. Didn't make a lot of playlists but the ones I made were accurate.	-	Yes, kind of. Don't know if it was from being in the gym or the playlist. Was mostly happier after listening. BUT a calm playlist made me very calm, it was very good.

5.4.4 Using MOODsic

The last part of the finishing interviews concerned the overall usage of the application and its user interface (see Table 5.9). All eight participants found the user interface to be user friendly and easy to navigate, where the majority highlighted that there weren't that many steps or interactions to complete in order to get a playlist. They also pointed out that there wasn't much content to consider and that they could find what they wanted, that it thereby was clear what to do.

When asked to expand on particular aspects of the interface, the participants had different viewpoints. Four of the participants pointed out positive aspects of the user design, such as the colors, the graph, the hover effects and the overall look. Two participants pointed out the buttons as particularly useful, where one highlighted that the buttons indicate the next steps of the experience and thereby confirms what they think happens next. One participant pointed out the fact that it is a web-based service instead of a mobile application, meaning the latter option would probably have led to more usage from them.

Table 5.9: Summarized answers about the application interface.

	User-friendly and easy-to-navigate user interface?	**Particularly useful or challenging aspects of interface**
P1	Needed to connect to Spotify each time I used it – bit of a hurdle. The rest makes sense, not much content. Can clearly see what mood to choose.	That it is a web service, would've helped if it was an app, like Shazam. If integrated with Spotify, I probably would have used it more.
P2	Very user friendly. No problem to find. Found where I wanted.	Nothing difficult, very easy to generate a playlist. Had no problem at all.
P3	Yes. You don't need too much interaction to generate a playlist, weren't that many steps.	Used mostly the standard buttons, they work well. You understand everything, wasn't hard to find things.
P4	Yes, it was clear what to do, not step by step, just one mission. Even though the boxes were different buttons, they still felt connected. Also liked the hover effect. And not hard interpreting the axes.	Good that the button says the next step, can focus only on picking mood. I know what will happen next. If I regret it, I feel like I can back out. You get confirmation beforehand. I liked it.
P5	Yes. Not much to misclick. Used it mostly on the phone.	Text on the cubes that say what mood it is. Can be difficult to understand on a small screen or if you're color blind.
P6	Yes. It was very good. Looked professional and well organized.	No, can't think of anything.
P7	Yes, I think so. It worked well.	At first – a bit difficult to interpret the graph and predict exactly what mood it would be. Didn't know how to think at first, but then it was no problem.
P8	Yes. I was already logged in to Spotify, so it was incredibly nice. Liked that the mood graph was visible on the whole screen without scrolling.,	Yes, liked that you can open the generated one in Spotify. Hardly used the "pick new mood" button – once you've created one, you don't want another.

When asked about certain features that they found difficult to use or understand, half of the participants found it simple and straightforward to use and mentioned no difficulties (see Table 5.10). The other half of the participants found the mood graph difficult to understand at first, particularly predicting what mood each box would be, but all said that they just had to think logically and learned how to interpret it after a while. Two participants mentioned the login-process to Spotify, where one found it to be a hurdle to have to connect every time and the other was constantly logged in and had no trouble at all.

Table 5.10: Summarized answers about the challenging features.

	Features that were difficult to use or understand	**How did you overcome these challenges?**
P1	No. You choose a mood, then get a playlist, not much to understand.	-
P2	No, if I haven't missed anything. The graph was very clear. The colors also fit, felt right. It was reasonable.	-
P3	At first you had to think what was what when choosing based on the color and the diagram. But you learned after making a few.	You just had to think logically.
P4	No, very simple to use.	-
P5	No. Maybe would have liked the mood picker right away, but understand why the start page is first.	Clicked the button.
P6	No. Maybe what box to pick, because it was just "a little less angry", but you figured it out.	You learn after a while.
P7	The graph, in that case.	A bit of testing in the beginning. But when the playlist is created, you get a picture of how it works.
P8	No, not at all. Not a lot of stuff you can do, pretty straightforward.	-

Additionally, the participants were asked to expand on ideas for usable features of the application that were not included (see Table 5.11). The overall theme was that the participants wanted more control of the outcome of the playlists, by choosing options such as which genre to base it on, the number of tracks in each playlist and if the list is based on their personal library, only new songs or a combination of both. Two participants mentioned that they would have liked to choose the look of the playlist in terms of cover image and playlist title, and one of these participants also mentioned being able to add tracks to already existing lists. Another aspect that emerged was integration with Spotify, as two participants thought that would make the experience smoother. Similarly, one participant proposed the idea of integrating it with Google Assistant, which uses voice control. Two of the participants could not think of any missing features.

Finally, six of the participants confirmed the possibility of them using the application in their everyday life beyond the study period. When asked why they would continue to use it, three of the participants pointed to its ability to provide music of a certain mood. Half of the participants said they would mainly use it to find new music beyond their existing playlists and known tracks, with one highlighting that it made the process of finding new, good music less tedious. One participant pointed out the application as a useful tool for getting a mix of their own music taste without having to make the playlists themselves, and another said that they would continue to use it since it worked well during this study period. The remaining two said they would use it only if it would have been integrated with Spotify, where one highlighted that they would want more control of the outcome themselves.

Table 5.11: Summarized answers about future use and features.

	Useful features that were not included	**Future use in everyday life**	**Incorporated in daily routine**
P1	- Integrate with Spotify - Control which genre you get	Probably not. If it had been part of Spotify I would have used it, but it's too much work to go into another service for the little I listen.	Every time you're in the mood for new music, you make a new playlist. If it had been in Spotify -- playlists updated instead of getting a long list of playlists.
P2	Not use your own history, nothing specific, just a mix. If you feel like exploring a bit.	Very possible. Spotify has the same ability, so it competes a bit. But yes, easy, would probably make a playlist again. To find new songs, so you take the best from the generated and put in your own.	Maybe when I'm going to game and listen to music, I'd make a new list. Then you make a new noe when you've listened through it.
P3	Design your own front image, or choose between how it would look, like an avatar/smiley depending on the mood.	Yes, 100%. Because it's a very good idea. I think it's hard to find new artists and songs that fit my music taste, with this you don't have to search Spotify for hours.	Don't think I would have used it everyday, but when probably when i listened through a list and tire of it. Every two weeks to get new songs and not listen to the same.
P4	- Not recommended songs, only songs you've listened to before - Choose the length of the playlist - Center it around a specific genre	Could imagine if more choices before creating, control the content yourself. That's the charm of a playlist, here you get a little help but you still want to contribute.	If got to choose myself -integrated with Spotify so it's a smoother experience. Or get some reminder, like in the morning. Or that Spotify notices when you need it.
P5	- In a real app, would have been nice - Voice control, say "Hey Google" and get into that mood	Yes. It worked well. It did its job, made me happier when I made a happy playlist. And easy to use.	If I'm cooking or doing the dishes, maybe not when running. But when doing something else at the same time.
P6	No, can't think of anything that could have changed.	Yes, I think so. Would like it when you want a little more music or inspiration, music you've never heard before and not the same old. And that you can choose the mood.	Would probably keep the lists I've created until I want a new one. Wouldn't create new everyday, only when I tire of an old one.
P7	- Rename the lists directly - Add to an existing list - Choose number of songs - Choose if it's based on only my own songs or new songs	Yes. Maybe not the way I first intended, but definitely to find other artists and partly if it's a certain mood or type of music I want.	If I listen in the morning, I could make a list to start the day on. And there were several times I created lists while cooking, when I just wanted a list to listen to.
P8	Don't know, not that I can think of. It has what it needs.	Yes. Especially at the gym. I'm not very good at making own playlists, with this I don't have to think about the content and just get a mix of my music taste.	Don't know. When I want to take it easy, because it was the best match for what I wanted. And at the gym. When studying, I didn't know what mood to choose.

6 Discussion

In this chapter the findings are discussed in relation to the research question and hypotheses. The research question for this study was as follows:

> **RQ:** How can a music playlist be generated, which is personalized according to a user's listening history and desired mood?

This was answered through the implementation of MOODsic, where a combination of technologies were used in conjunction with affective studies to develop a playlist generator based on desired mood and individual listening data (see Section 4.1). With the use of this tool, playlists could be generated by connecting the valence-arousal levels of the desired mood, according to the dimensions presented by Russell (1980), to the corresponding mood-related audio features of each track, provided by Spotify's API. Each playlist was personalized by distorting the range of the features and basing the recommendations on previous listening data. In this way, both the user's affective context and their user preferences were taken into consideration.

The application could then be evaluated in order to be able to test the two hypotheses:

> **H1**: Users will confirm that the generated playlists matched their desired mood.
> **H2:** Users will confirm that the application that generates such playlists is engaging.

First and foremost, the findings showed that the participants seemed to incorporate the use of MOODsic into their everyday contexts, though it was not used every day of the study. Each stage of the data collection points to participants creating the most amount of playlists in the beginning of the testing period, with participants even reporting this themselves. The interviews revealed that the reason for this might be that some of the participants were initially unsure about how to interpret and navigate the mood graph and were testing it out in the beginning by experimenting with the creation of playlists.

A theme that emerged was to use the application while conducting everyday tasks, which corresponds to the answers of the pre-study relating to usual listening habits (see Section 5.1.1). In addition, participants underlined that music was not a tool needed for survival but that it plays a part in changing or improving one's mood, which agrees with previous research in terms of music listening motives (Getz et al., 2014; Schäfer et al., 2013; Putzke et al., 2014).

For some of the participants, the approach to creating a playlist became dynamic in that they created one new playlist for each listening session. Similar to Hagen's

(2015) findings, this shows that playlists were used to assemble fleeting preferences, and in this case affective states. However, these playlists became static over time in that some participants reported only listening to each playlist once and then creating a new one when needed.

The participants self-reported to have listened about the same amount throughout the period, which the interviews established as well, with participants underlining that the amount of music they listened to was not more than usual. However, most of the participants listened more to the generated playlists than their own, or an equal amount. Regardless, the majority of the participants found themselves listening to more unknown songs and exploring new artists, an aspect which was not initially expected by the use of the application. Overall, all of the participants found the experience of discovering new music to be a positive one, and some added occasional tracks to their personal libraries. In addition to finding new music, some participants rediscovered tracks or artists that they hadn't interacted with in a while, which also had a positive effect on the experience. This correlates to Goethem and Sloboda's (2011) findings on music's role in evoking a certain nostalgia.

In regards to familiarity, all participants found that the playlists were accurate to their music taste, which was to be expected considering that the recommendations were based on their top listening data. While some participants reported having more new content than familiar in their generated playlists, most said that the recommended songs still matched the style of the others. In other cases, participants said that some tracks didn't fit in at all. This could be connected to the playlist data in terms of the difference between target and mean, where some of those playlists may be the recognized outliers (see Figure 5.13). Overall, the mean difference between target and mean in each playlist was less than 0.01, indicating that the recommender system is accurate in matching the style

When comparing the generated playlists to the desired mood which they were adapted to, opinions varied on the accuracy of those lists. The self-reports showed that most of the participants found the generated playlists to be accurate in terms of their desired mood throughout the study period. However, when interviewed about this, only half of the participants reported the playlists to be accurate. The mood which most participants pointed to as being especially inaccurate was angry, which doesn't correspond to the spread of the angry playlists as shown in the plots.

The spread of the tracks in each playlist differed for each participant, where some had compact playlist spreads and others had scattered ones throughout (see Section 5.3.3). However, playlists that were based on calm or happy moods were more spread than others, regardless of participant. One reason for this might be the nature of the participants' music taste, as the genres of music which the participants reported to prefer were of a wide range. The participants that classified their music taste as narrow resulted in more compact playlists overall in terms of valence and

energy, but the participants with broad music taste had a more varied spread in their playlists.

The narrow nature of the participants' music taste might have resulted in additional limitations for the generator. One theme that emerged in the playlists was that when choosing a specific mood, some participants only got one type of music (e.g. lo-fi for sad, rap for angry). This limitation in their music preferences is an aspect that some of the participants pointed out themselves. In contrast, some of the participants' playlists may have been affected by their broad music taste in that certain genres or artists were missing from their playlists.

Additionally, one participant pointed out that they judged the new songs in their lists even before listening, indicating that the aspect of self-selected music may have had an impact. This participant also highlighted the want for more control of the outcome of the playlist, further strengthening the idea that personalized music has the potential for a stronger affective experience (Baltazar et al, 2019; Goethem & Sloboda, 2011, Papinczak et al., 2015). In addition, they saw the creation of playlists as a form of self-expression, similar to Hagen's (2015) findings.

As the prototype was developed with affect regulation in mind, one aspect to consider was if the participant ever experienced a change in mood while listening to these generated playlists. The pre-study presented that music was found to be an important tool for changing the participants' moods, but that they mostly wanted to change it for the better. The playlist data demonstrated this in that the majority of the created playlists were either of the happy or calm type. In addition, the interviews revealed that some of the participants changed their mood for the better after listening to one of the playlists with higher valence. Some of the participants reported the use of the tool to enhance their existing mood instead of generating new ones, which correlates to Siles et al.'s (2019) findings. The pre-study also highlighted that some of the participants often did not want to get mad, which could be an explanation to why angry was the least picked mood.

Another theme that emerged through some of the participants was the aspect of taking the contents of the tracks into consideration when classifying music according to mood. For sad music in particular, several participants pointed out that the mere musicality of a track does cultivate that mood and that the message should also be considered. As mentioned in both interviews, some participants deemed this aspect missing from their generated playlists.

Regardless of the mood chosen, the majority of the participants reported that they would continue to use the application in their everyday life if possible, though a few would need it to be integrated directly with Spotify. However, the primary reasons behind the participants' continued usage might not be to get personalized playlists based on desired mood only, but also to find new, relevant music that has the potential to bring some inspiration into their daily lives. They highlighted the

usefulness of the tool in terms of not having to search for and gather new music themselves and thereby simplifying the experience. The overall experience was found to be enjoyable, as multiple participants pointed out the lack of interactions needed in order to get a final playlist. This was considered an important aspect of the prototype, as it was meant to put emphasis primarily on the experience of generating a playlist based on specific criteria. The main difficulty that emerged was interpreting the mood graph in initial stages, but after continued use the participants figured it out.

With all of this in mind, both of the hypotheses can be accepted; users confirmed that the generated playlists matched their desired mood, with a few playlist outliers, and that they found the application as a whole to be engaging. However, the aforementioned aspects, such as the lyrics of a track, or missing artists, or whether a track is familiar or not, can all be deemed subjective areas of the listening experience. Each of the participants had a different, personal experience in using this application in their everyday life. Given this individualized aspect of music listening, there are some factors that the playlist generator was unable to take into consideration.

7 Conclusions and Future Work

7.1 Summary

To conclude, this project investigated the topic of creating personalized playlists based on individual listening data and desired mood. The research question was answered through the implementation of the playlist generator MOODsic, which in turn was tested on users in order to test the given hypotheses. The results showed a general indication that the playlists were accurate to the participants' musical preferences, and with both new and familiar content in the lists. Most of the participants reported that the majority of their playlists matched their desired mood, though in some cases specific songs or moods stuck out as being particularly inaccurate. For some participants, outliers occurred in terms of the difference between target and mean energy and valence, meaning that some tracks did not match the intended target of the playlists. This could be explained by the individuality of participants and the wide range of their musical preferences. As a whole, these results hint at the personalized aspect of music listening as being particularly important when addressing affect.

In addition, the findings indicate that participants incorporated the application in their everyday life and mainly used it as aid in everyday tasks. In terms of affect regulation, the application was mostly used to enhance the participants' moods for the better or enhance their existing moods. Overall they listened the same amount throughout, and more to the generated ones than their own, but generated more playlists in the beginning period due to initial confusions regarding the interface. The application as a whole was found to be mostly useful in terms of finding new, relevant music and thereby facilitating the search for personalized music recommendations. In other words, both hypotheses could be accepted, meaning that the users both confirmed that the generated playlists matched their mood and that the given application was found to be engaging.

7.2 Future Work

In order to make the generated playlists more accurate in terms of both desired mood and music preferences, a few changes could be implemented in such a recommender system. First of all, the other available mood-related features that Spotify provides could be used to further classify the music based on mood. These features include tempo, danceeability and instrumentalness, but could also include other characteristics. However, as the research behind the use of these features were lacking, no concrete way of using them was found without going beyond the scope of research. Additionally, the mood graph could make use of the original ranges (0-1) instead of the distorted ones when classifying the desired moods of the users, or compare the two approaches to each other. This could potentially make the moods and corresponding generated playlists less individualized, but therein lies the

possibility of gratifying the recommendation needs of the participants whose music preferences was a possible limitation. In addition, a few of the options pointed out by the participants could be implemented, such as control over additional factors including the genre of music, the length of playlists and the integration with Spotify.

Future work in this area could also make extended use of ESMs, where self-reported data could be collected at multiple times per day instead of per week. Similarly, daily or hourly changes in affective states could be monitored in order to compare them with the generated playlists and their corresponding listening data. Disregarding the limitations of retrieving past listening history from Spotify's API, additional research could then make use of users' listening data in comparison to the generated playlist to investigate and point out patterns in music listening when regarding its relationship to affect and individual preferences.

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Appendices

A Pre-study questions

Music Preferences

1. Could you tell me a little about your music listening habits?
    a. How frequently do you listen to music?
    b. How important is music in your daily routine?
2. How would you describe your music taste?
    a. Do you prefer specific genres or artists?
    b. What genre(s) of music do you enjoy the most?
    c. Would you say your music taste is broad or narrow?
3. What type of music do you listen to when you are working/studying?
4. What type of music do you listen to in your free time?

Moods and Music

1. How important is music to your overall mood and well-being?
2. How would you typically describe your mood when listening to music?
3. Have you ever experienced a change in mood while listening to music?
4. How do you typically choose music to listen to when you want to change your mood?
5. What kind of music do you listen to when you want to feel…
    a. Happy?
    b. Calm?
    c. Sad?
    d. Angry?
6. Do you currently use any mobile applications related to music? Have you ever used one?
7. Do you currently use any mobile applications related to mood? Have you ever used one?

Listening Habits

1. What platforms do you use to listen to music? (e.g. Apple Music, Spotify, etc.)
2. How do you typically interact with your mobile device while listening to music?
3. When you listen to music, do you use playlists or you listen to albums and individual songs?
4. Do you make your own playlists or only subscribe to others?

B Self-report questionnaire

Moodsic Questionnaire

Time to think back on the usage of Moodsic so far!

* Indicates required question

1. Name *

2. Have you generated any playlists these past few days? *

Mark only one oval.

- I have not generated any playlists
- I have generated a few playlists
- I have generated some playlists
- I have generated many playlists
- I have generated a large number of playlists

3. Have you listened to any playlists these past few days? *

Mark only one oval.

- I have not listened to any playlists
- I have listened to a few playlists
- I have listened to some playlists
- I have listened to many playlists
- I have listened to a large number of playlists

4. How accurate do you think the playlists were to the desired mood? *

Mark only one oval.

- Very inaccurate
- Somewhat inaccurate
- Neutral
- Somewhat accurate
- Very accurate
- Cannot answer -- I haven't listened to any playlists

5. Is there something about the experience you would like to elaborate on?

C Finishing interview questions

Listening patterns

1. How often did you use the application during the two weeks of the study?
    a. How often did you make playlists?
    b. How often did you listen to playlists?
2. Can you describe your typical usage patterns?
    a. For example, did you use it more in the morning or in the evening?
    b. Did you use it on weekdays or weekends?
    c. Did you use it while studying/working or in your free time? Or both?
3. How did the application impact your music listening habits during the two weeks of the study?
    a. Did you find yourself listening to more music than usual, or exploring new genres or artists?
    b. Did you listen to the generated playlists more often than your own playlists or other music? Or was it the other way around?

Generated playlists

1. Did you find the personalized playlists generated by the application to be accurate and reflective of your music taste?
    a. Were there any playlists that you found particularly accurate or inaccurate?
2. Were there any genres, artists or individual songs that you felt were missing from the playlists?
    a. Were there any specific genres or artists that you would like the application to exclude?
    b. How important is it for you to have a wide variety of artists and genres represented in your playlists?
3. How familiar did you find the songs and artists in your generated playlists to be?
4. Playlists may have included songs that were not in your recent listening history. How appropriate did you find the inclusion of these songs?
5. Did you discover any new songs or artists through the playlists generated by the application?
    a. How did you feel about these discoveries?
    b. Did you add any of these songs or artists to your personal music library or playlists?

Mood

1. Over the course of the study, how reflective did you find the generated playlists to be of your desired mood?
2. Were there any instances where the application generated a playlist that was not reflective of your desired mood?
    a. Can you describe the situation and how the playlist differed from your desired mood?
    b. How did you feel about the playlist in this situation?

3. Did you ever experience a change in mood while or after listening to a generated playlists?

Application usage

1. Did you find the user interface of the application to be user-friendly and easy to navigate?
    a. Were there any particular aspects of the interface that you found useful or challenging?
2. Were there any features that you found difficult to use or understand?
    a. How did you overcome these challenges?
3. Were there any features that you think would be useful but were not included?
    a. Do you have any other suggestions for improving the user interface?
4. Would you continue to use the application in your everyday life beyond the study period?
    a. Why or why not?
    b. Can you describe how you might incorporate the application into your daily routine?

D Moods dataset

Main mood	Color	Hex	Valence Description	Energy Description	Valence Feature	Energy Feature
angry		#d63e3e	low	high	0.2	0.8
angry		#e85f34	semi-low	high	0.4	0.8
angry		#db3c5d	low	semi-high	0.2	0.6
angry		#e35c53	semi-low	semi-high	0.4	0.6
happy		#fbb416	semi-high	high	0.6	0.8
happy		#f7d720	high	high	0.8	0.8
happy		#e8c632	semi-high	semi-high	0.6	0.6
happy		#dad815	high	semi-high	0.8	0.6
calm		#78d454	semi-high	semi-low	0.6	0.4
calm		#98d825	high	semi-low	0.8	0.4
calm		#00d465	semi-high	low	0.6	0.2
calm		#5ad63e	high	low	0.8	0.2
sad		#6a8ad9	low	semi-low	0.2	0.4
sad		#54a2de	semi-low	semi-low	0.4	0.4
sad		#3e97d6	low	low	0.2	0.2
sad		#04b2dd	semi-low	low	0.4	0.2

E Pre-study answers

E.1 Music Preferences

Participant	Could you tell me a little about your music listening habits?	How frequently do you listen to music?	How important is music in your daily routine?
1	Always when walking to school, at the gym, on a walk, when cleaning.	Every day.	Could survive without it, but always have music on when going somewhere.
2	All the time - when cooking, going to school at school, at the gym, in the shower, when gaming.	Every day, 4-10h/day. Always has music on.	Very important. Without it, much would be difficult. Makes it easier to focus.
3	Mostly when at the gym, or home cleaning. In the gym every day, so listens every day.	Every day, basically.	Wouldn't be able to go to the gym without music, so quite important. But in other contexts -- not that important, rather listens to podcasts.
4	Usually listens to music while doing something else at the same time. Like when working, doing holsehold chores (like doing the dishes), when getting ready go to go out or when driving.	Every day.	Very important. Has to get the music going before starting on some stuff. Like before going to work, or before getting to a specific task at work. Turn on the music, then you're up and running.
5	Mostly when cooking or driving. Mostly to express feelings or get some kind of nostalgia.	At least once a day. Usually twice. In sessions.	Not that I can't function without it, but it makes a lot of things more fun, so pretty important.
6	Usually in the morning, when I work out, when I'm cooking, sometimes when I'm sitting at school. It's almost 24/7.	Mostly every day. Somedays it's podcasts instead.	Very important. Otherwise it will be very gloomy and quiet.
7	Listens a lot at home, on a shared speaker (so not my Spotify). Otherwise it's when I'm going somewhere, taking the bus or going to school, or when studying.	Every day. Although it's a big difference from days when you have it on in the background or if you only listen sometimes.	Yes it's important, I guess. Sometimes it's nice to NOT listen to music too, if the brain is overloaded. But sometimes it's nice with music.
8	Yes, I listen to music on Spotify every day. Every time I walk to school, when I cook, when I clean, when I exercise, when I study. Listens almost all the time, except when having a conversation.	Every day. When I'm at school, it's probably 5-6 hours a day.	Very. It's not something I have to have, but it's nice to have available.

Participant	How would you describe your music taste?	Do you prefer specific genres or artists?	What genre(s) of music do you enjoy the most?	Would you say your music taste is broad or narrow?
1	A bit of an omnivore.	Mostly rap, a little rock. When I work out, it's often a bit heavier music — metal, when I run — techno, house.	Same as previous.	Likes a lot and can listen to everything, but most listened to tracks is in narrower category.

2	Very mixed.	House, lo-fi (study music) melodies and nice beats. Hardstyle. Also listens to radio music, top charts.	Depends on the situation. At the gym - hardstyule, when I'm studying - lo-fi.	Broad, very broad.
3	A lot of Swedish rap and 90s R&B and rap.	Not reallly, just those just mentioned. Also lo-fi beats and Swedish R&B and pop music.	Same as previous.	Very narrow.
4	Extremely varied. But definitely more uptempo music in all its forms, rhythmic. Don't like listening to very calm songs. Of course sometimes when you need to unwind. And when you have to put on some music at work that others can listen to. But rhythmic, uptempo music.	Love pop music, and also rock and R&B in all its form, and disco funk. Very decade-bound. Has lists for the different decades.	See previous answer.	Very broad, obviously. Classical music is also a favorite.
5	Listens to everything.	Mostly indie rock or old hip hop.	Indie rock and hip hop.	Very broad.
6	Likes 90s, 00s R&B best when listening at home. At the gym -- mostly hard music, a mix of electronic music, techno, psytrance.	I like like Rihanna, Eminem. Genre - R&B, Reggae, Pop.	See previous answer.	In between -- wouldn't say it's wide. Depends on what I do in everyday life. When I have people at home it's calm music, when I'm working out it's upbeat music, and when I'm at home by myself it's somewhere in between.
7	Depends on what mood I'm in. Tends to switch between different kinds of music. If calm songs -- indre rock music. If excited -- more rap or rock.	It sort of goes in periods. But I probably oscillate mostly between rock, rap and a bit of R&B. Usually don't think about genre -- If I like a song, I like a song. Depends more on mood or what period you're in than what genre or artist it is.	See previous answer.	Broad, I guess.
8	Broad.	Pop and rock, I guess. And rap sometimes, but not very often.	Rock.	Broad.

Participant	**What type of music do you listen to when you are working/studying?**	**What type of music do you listen to in your free time?**
1	Either the usual (rap) or film soundtracks, since there’s no lyrics to disturb.	Often rap.
2	Lo-fi.	Radiomusic, popular music.
3	Probably lo-fi.	A lot of Swedish rap, R&B. Regular rap and hip hop top.

4	Usually very uptempo, that makes you want to dance. Can't listen to calm music at work. Usually puts on big playlists of all liked songs.	Then its more albums, but also lists of course. But mostly albums that you listen to from start to finish.
5	Everything possible, not that specific. Mostly songs where not too much happens, but not too relaxing music either.	About the same, but then it's really everything, including hype music.
6	Then I have a radio-list that updates by itself, based on the song Angel by Shaggy, that changes a little all the time. It's kind of R&B.	Kind of the same. I don't have a list, but I usually search if there is something, then it finds its own songs.
7	Depends on whether I want to concentrate or not, I can't always focus when I'm listening to music. If I want to focus -- a little less singing and more sound, but it doesn't have to be quieter.	Everything else.
8	Can actually vary a bit. Mainly lo-fi, but sometimes rock.	Rock, in that case.

E.2 Mood and Music

Participant	How important is music to your overall mood and well-being?	How would you typically describe your mood when listening to music?	Have you ever experienced a change in mood while listening to music?	How do you typically choose music to listen to when you want to change your mood?
1	No large affect. Listens because he likes music, not to cry or something. Uses it when working out, however, in order to get hyped.	Both when I'm happy, sad, angry. No specific mood.	Yes, would probably say that. Can get into a certain mindset, get hyped, when listening to a really rough rap song.	Sometimes when listening to something and feeling like it's too much, you switch to something calmer. Same in the evenings. But if going out and doing something, it might be something else.
2	Very important. Can easily change one’s mood.	Mostly happy, sometimes sad, and hyped if you’re going out.	Yes. If you’re tired or at the gym, could listen to energetic music to get excited and get energy.	Have a few different playlists. Or search for a playlist that describes my mood, e.g. "Good mood playlist".
3	When at the gym, it helps to be more focused on what you have to do and shut out everything else. Same when you're studying. So pretty important.	You become more focused, which makes you more productive. Contributes to getting something done.	Maybe if you've been dumped and listened to sad music, you become ready to cry. But somehow you feel better when listening to music and getting something done.	Usually uses playlists on Spotify and get new ones that they think fit into my playlist, and also use the Radio feature.
4	Very important. Wouldn't say I depend music for my mood, but it enhances the mood I'm in. So it's important to enhance feelings or mood.	Happy. Or motivated, excited.	Yes, yes. For the moment, I would say. Not for long.	Usually wants something uplifting, rarely drawn to music that makes you down. Haven't listened to sad list in ages. Also like songs where you can sing along, then calm songs don't go so well.

5	Fairly important. It affects quite a lot. If you're having a bad day, you can listen to a good song and it turns your mood around. And vice versa.	Generally good, enjoyment.	Yes. It can be anything from being sad to happy or sad to happy. Or e.g. if you run, you can listen to something that makes you energetic. Same when I drive.	Most of the time it's music I know makes me feel good or feel better. While broad, I am very picky in what I listen to and skips if it doesn't fit into what I'm feeling at the time.
6	I think it affects a lot. Time slows down if I don't listen to music. It's much harder to get up in the morning without music, and same when going to school or driving. Becomes very boring without music.	All moods. But overall -- like I'm doing well. Probably listen more when I feel look than when I feel bad.	Yes, definitely. If it's a sad song you can get song, but if it's a bitchy song, "f you" song, you can get in that completely different mood.	Depends on my mood and what I want to change it too. If I'm getting hyped for the gym -- electronic music, hyped for a party - old party songs. Most of the time you don't want to become sad, but if you would, then some sad song. And if I want to be happy, I like to put on a reggae or R&B song.
7	Sometimes I catch myself becoming happy after listening to calm or peppy music. Otherwise I don't think about why I listen to songs, I just put them on.	Either I'm just excited and want to dance, it doesn't matter the music. I listen to songs I think are good.	Yes absolutely. Maybe if it's just plain music and I turn on peppy and upbeat music, that makes me hyped.	I don't think I want to change my mood. Don't always listen to songs that match how I'm feeling, just think of an artist and put it on. Probably change my mood automatically, without being an active choice. Sometimes you feel that the music doesn't match, then you change it.
8	The music varies depedning on the mood. It represents my mood.	Usually happy or gloomy, those are the two biggest ones.	Yes, I have. When I'm cooking -- happy music, when I'm studying -- more somber music.	I usually use mood as a keyword when searching on Spotify, like "Happy" or "Upbeat" and "Slow" when you're sad.

Participant	**What kind of music do you listen to when you want to feel…**			
	Happy?	**Calm?**	**Sad?**	**Angry?**
1	House, and old songs. Songs you can sing along to.	Songs without much text, more melodic. Piano music, movie soundtracks. Where you can zone out a bit.	Depends on what makes you sad — difference in if someone's passed or if its hearbreak. Music where you can relate to the sadness.	Hard metal or heavy rap, so you can get fully hyped.

2	Songs that represent fond memories, old favorite songs. Very mixed, no specific genre.	Lo-fi. Gets you concentrated.	Calm songs, like Lewis Capaldi. Has a list for calm, nice songs.	Don't want to get mad, has never listened to music to get angry. But IF -- probably hard rock.
3	Probably upbeat songs, pop. Kind of Swedish pop, new uptempo releases. Not ballads.	Then I listen to a lot of lo-fi beats, calm music, not so harsh.	Then I can listen to sentimental or old music.	Then I listen to hard hip-hop. Like Post Malone, Drake.
4	Festive music, and not just party music. Have a list with songs where you feel like the artists are smiling when they sing them. And also music that makes you feel empowered.	I would say R&B, old R&B above all. Also classical music sometimes.	Songs that remind me of past tragedies in my life that I can feel like "this one was written about me", whether it's about love or family. Like tragic songs.	Rap, like Tupac. Some music like that.
5	Nostalgic music that you've listened to before, road trip music type.	Kind of the same, or Dutch and German music. Nostalgic music.	Indie alternative type.	I don't often listen to music to GET angry, but if it was anything I guess it's heavy metal or death metal. Although it is very rarely, almost never.
6	Reggae, I would say. Party music if I want to get festive.	Probably James Arthur.	Like [James Arthur] then too. Or maybe Lewis Capaldi too, he has a lot of sad songs.	Then it's "f you" music. It depends on what it is I want to be angry about. But if it's a guy, then maybe it's Rihanna.
7	Either tempo songs that makes you want to dance, or nostalgic songs from when you were younger. Those you can sing along to. Don't have to be happy songs, as long as you know them.	Kind of the same, though not upbeat songs but songs that feel familiar and that you have listened to many times. Instrumental songs too.	I have a list for that with many songs on it, but generally pretty depressing. Doesn't have to be sad music, I go more for lyrics and am moved more by that than by the music.	I probably don't listen to music when I'm angry, I just sit and think. Anger probably doesn't match my music taste. Then I would listen to bad music I don't like, Country maybe. If I have periods where I listen to metal rock and I have been feeling bad, it kind of calms me down. Some songs make you feel sad instead.
8	Pop, I guess.	Lo-fi or classical music.	Not often that I want to be sad, don't actually know what to choose then.	Rock or hard rock. Sometimes rap, depending on the artist.

Participant	**Do you currently use any mobile applications related to music? Have you ever used one?**	**Do you currently use any mobile applications related to mood? Have you ever used one?**
1	Finds a lot of new music on Instagram. Shazam if you hear something good.	No and no.
2	Used YouTube to watch music videos, but not to find songs.	No, just entertainment apps and games which are there to put you in a good mood. And health apps and sleep apps. They kind of affect the mood

		indirectly.
3	Don't think so. Soundcloud and iTunes when you were little	No. Just a training app where you can fill in mood after each session, it's probably the most similar if you're going to track your mood.
4	Uses Genius a lot, but no others.	Checks Co-star every day, though not in detal. It's a bit about mood. But you don't set the mood yourself, it's more controlled by them.
5	Soundcloud at some point maybe, and also Garageband where you fiddle with your own songs.	No, but have probably done at some point. Study apps, but maybe that's not mood related.
6	Soundcloud, YouTube. Shazam many years ago.	No.
7	Shazam maybe, though I don't use it very much. There is music in almost all apps, but not where that is the purpose.	No, don't think so.
8	Shazam, but that's probably it.	No and no.

E.3 Listening Ways

Participant	What platforms do you use to listen to music? (e.g. Apple Music, Spotify, etc.)	How do you typically interact with your mobile device while listening to music?	When you listen to music, do you use playlists or you listen to albums and individual songs?	Do you make your own playlists or only subscribe to others?
1	Spotify and Youtube, most often. Sometimes Soundcloud to find remixes.	Changes songs often, has the mobile in hand in order to change volume and songs immediately. Never listens to songs all the way through. Every song is personally chosen.	Playlists. Reasonably sorted, have a few different ones for different things.	Mostly own playlists. Or Spotify's Daily mixes, to find recommended for me. Or using Radio on a song or a playlist, to find something new.
2	Spotify, SoundClound and YouTube.	Usually puts on a playlists, queues some songs from other lists, then just lets the playlist run.	Mostly playlists. It happens that you go into albums, songs and artists specifically, especially if you found something new.	Mostly own playlists, but not only.
3	Spotify.	I choose a list. I have a lot of lists based on what I want to listen to, what mood I'm in. Then I just let it go. I rarely choose songs, I mostly use Shuffle.	Mostly playlists.	I make my own. Always.
4	Spotify. Sometimes YouTube.	Looks at the lyrics a lot on Genius and keep up with them, to see the meaning of the texts. Then I queue when I want to listen, because even if you are on a list there are songs on other	Mostly playlists, but still a large proportion of albums.	When I listen by myself, I mostly listen to my own. In the office, I often listen to others, such as ready-made ones on Spotify. What I do a lot -- search for a

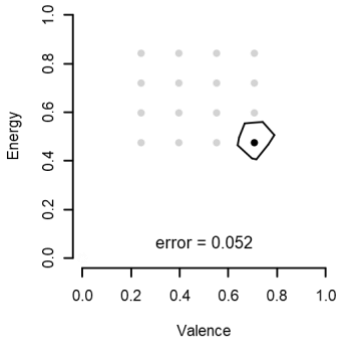
		lists that you want to listen to. Skips songs a lot, if i don't like the one that comes on.		song I like and click on playlists to find playlists based on that songs, with songs that have the same vibe.
5	Spotify. YouTube.	Usually leaves it be and just shuffle. But in certain moods -- quite picky, so then I can choose and queue. But mostly shuffle, tries to put it aside to not fiddle with it.	Playlists. Which I'm getting tired of.	Both. Lately it has changed that I listen to others -- they are better at organizing everything. But usually listen to my own.
6	Spotify, SoundCloud and Youtube. In that order.	Goes in quite often. Especially at the gym, then I go in and change songs. Try to let it go but have to skip if a song comes on that you don't like.	Usually playlists but not my own. If I have my own, I usually get tired of it after a while. Then I usually turn on the radio to a song that I think suits my mood.	More on others, which are updated. But then it will be the kind that are created for me.
7	Spotify.	Usually just put something on, been a long since queueing. If I want to find new music, I use radio on a playlist or song.	Mostly playlist, but also if you're on an artist page and plays those songs.	Own playlists. Maybe put on a general playlist when I work out. Sometimes I listen to Spotify's generated playlists, like "for you". But that is based on what you yourself have listened to.
8	Spotify.	These days it's shuffle on own playlists. But if it's something new, like an artist or playlist, then I just click play. If I find a song that I like then i queue it so it comes first.	Playlists. Either the ones I've created myself or the "This is..." type. And the ones that Spotify created for me, like daily mixes.	I have some playlists that other people made, and also shared playlists with friends. But otherwise it mostly my own and those made by Spotify, which are made for me.

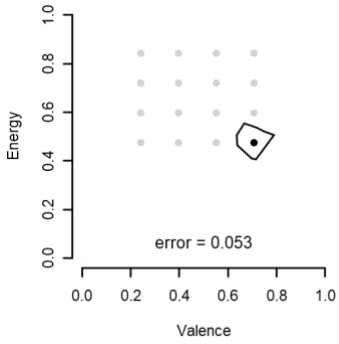
F Self-report questionnaire – Answers to open question

Participant	Round 1	Round 2	Round 3
1	I basically only listen to music on my mobile, a bit clumsy to have to make the list on the computer	-	Would have used more if it was integrated into Spotify
2	I think it was fun to get unique lists of both new and old songs!	-	-
3	-	Love it when sleeping and working out!	-
4	It feels like some songs fit perfectly in the list and other songs you don't think belong there at all	One of the latest ones I made (low energy & high happiness) had the best selection of songs so far. Appreciate how simple and quick it is to use. If one were to add more advanced settings maybe one could be that you could pick if you want more recommended new songs or only songs you have saved. Sometimes you're in the mood to only listen to your own songs.	-
5	It seems like sometimes the playlist forgets the mood it was created for and throws in songs that the listener likes but do not actually convey the mood that the playlist was created for, although this happens maybe 1/8 songs		
7	-	-	Haven't been listening to much music lately, just listening to podcasts. The last lists I generated didn't suit my mood, so maybe I was a little unexcited to make more

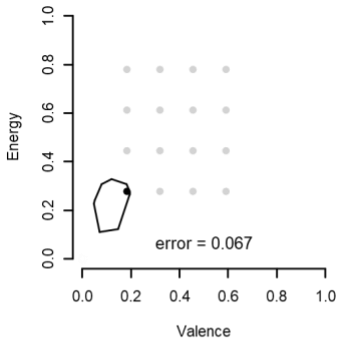
G Independent visualization of each playlist per participant

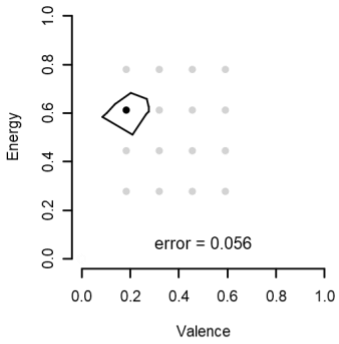
G.1 Participant 1

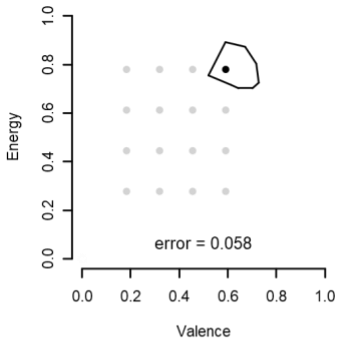


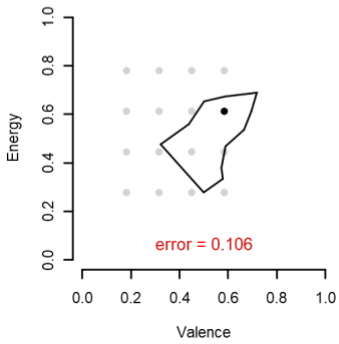


G.2 Participant 2









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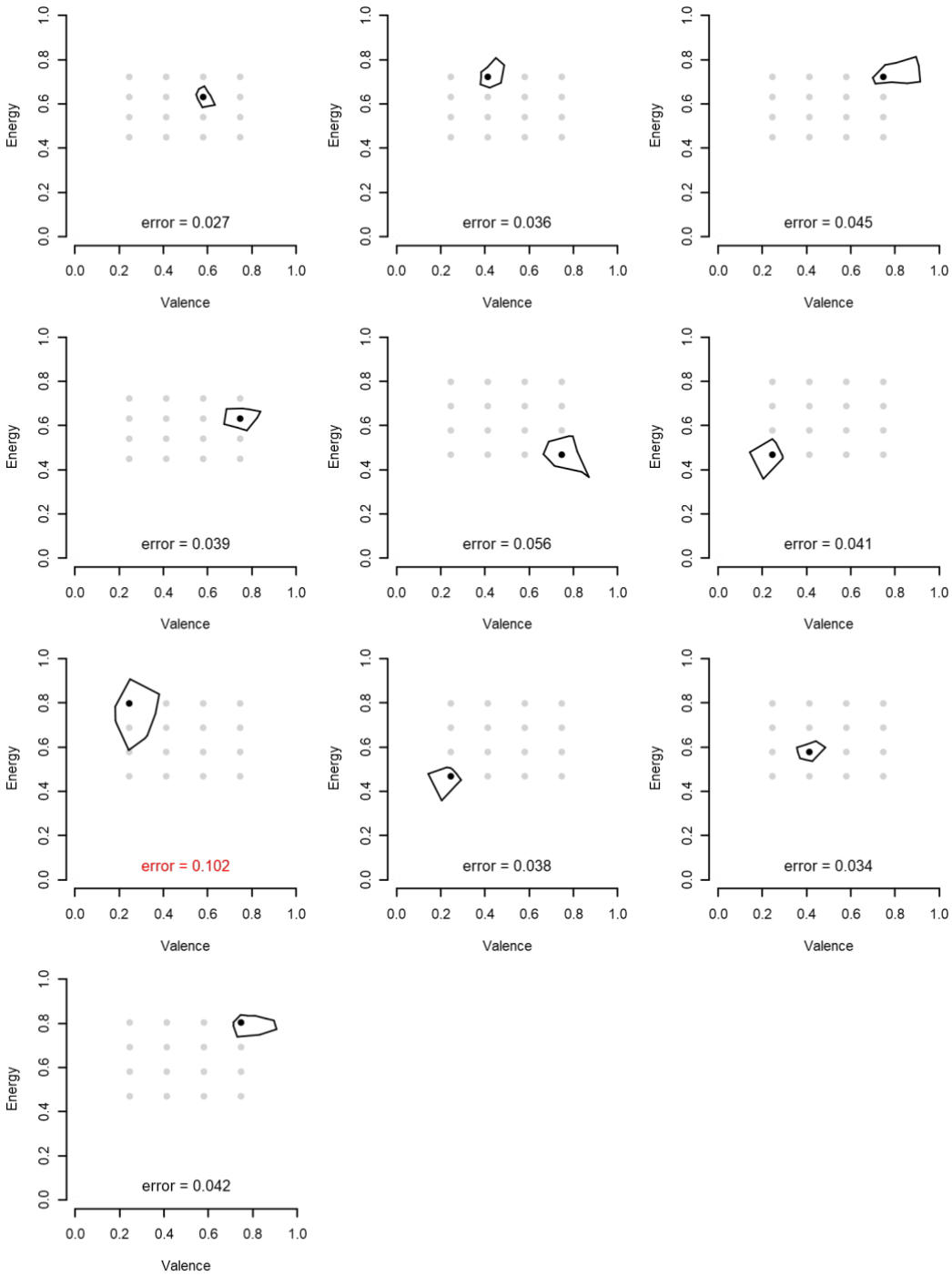
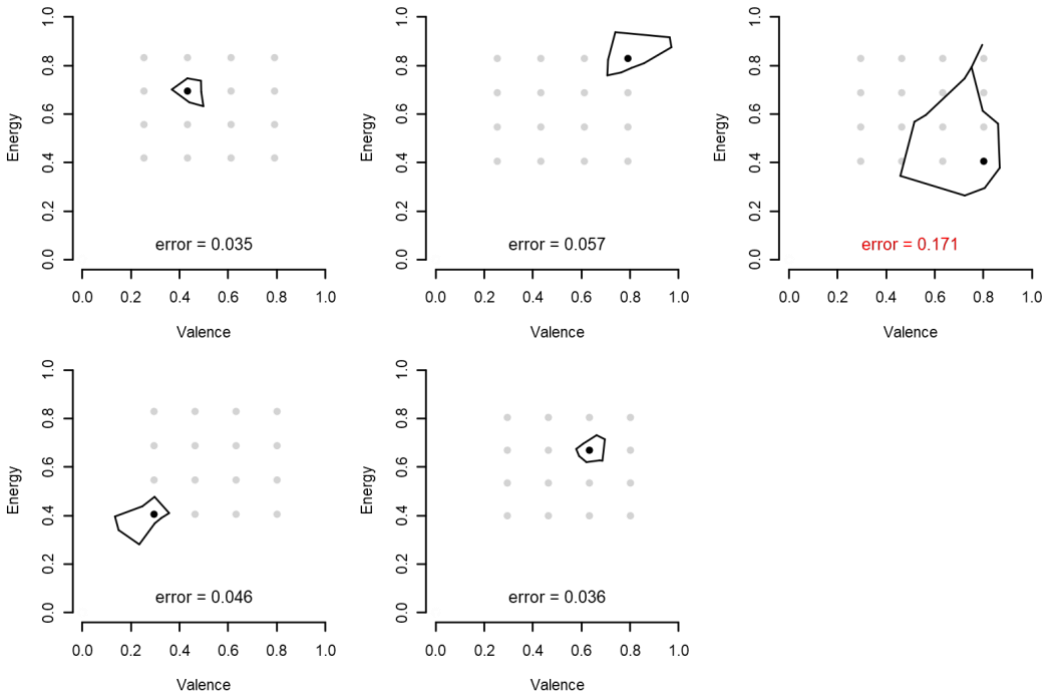


G.4 Participant 4



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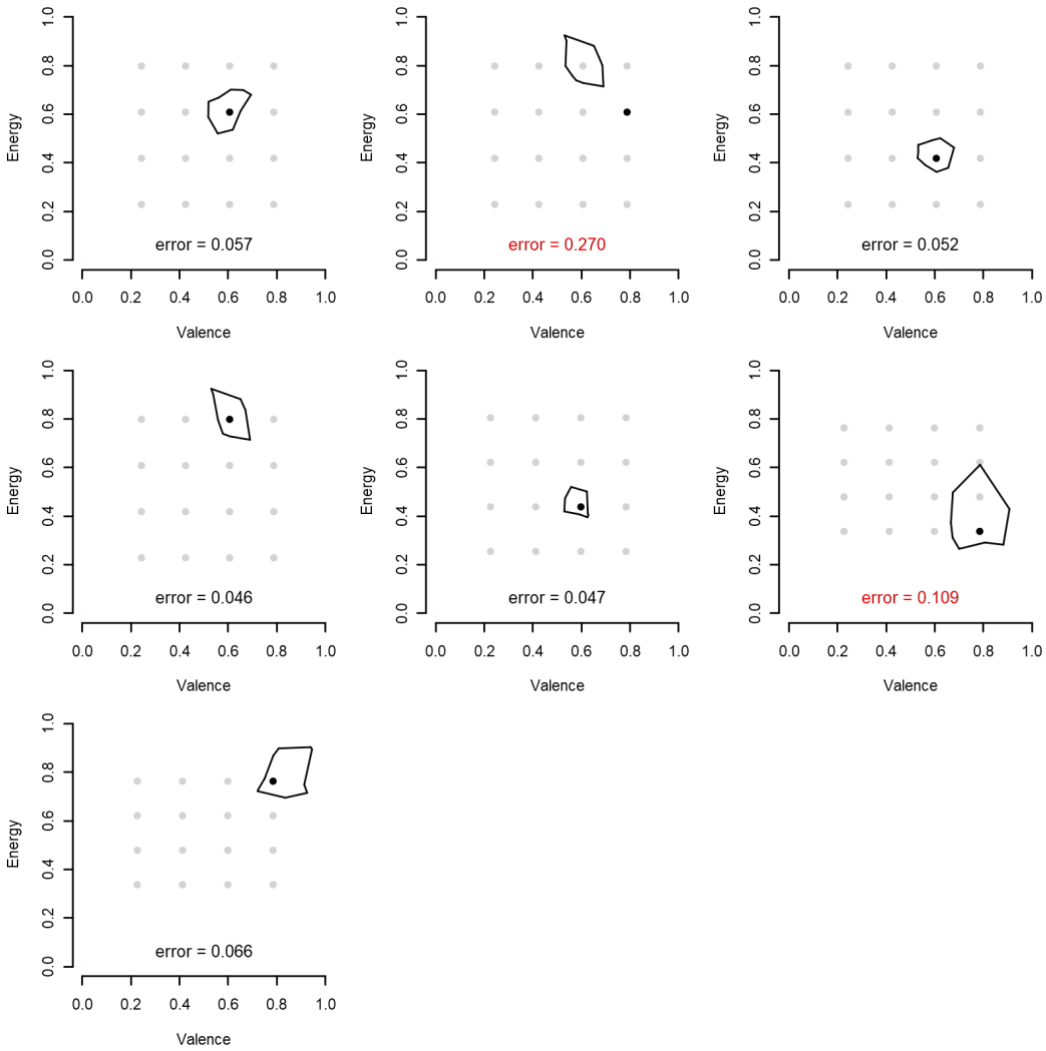

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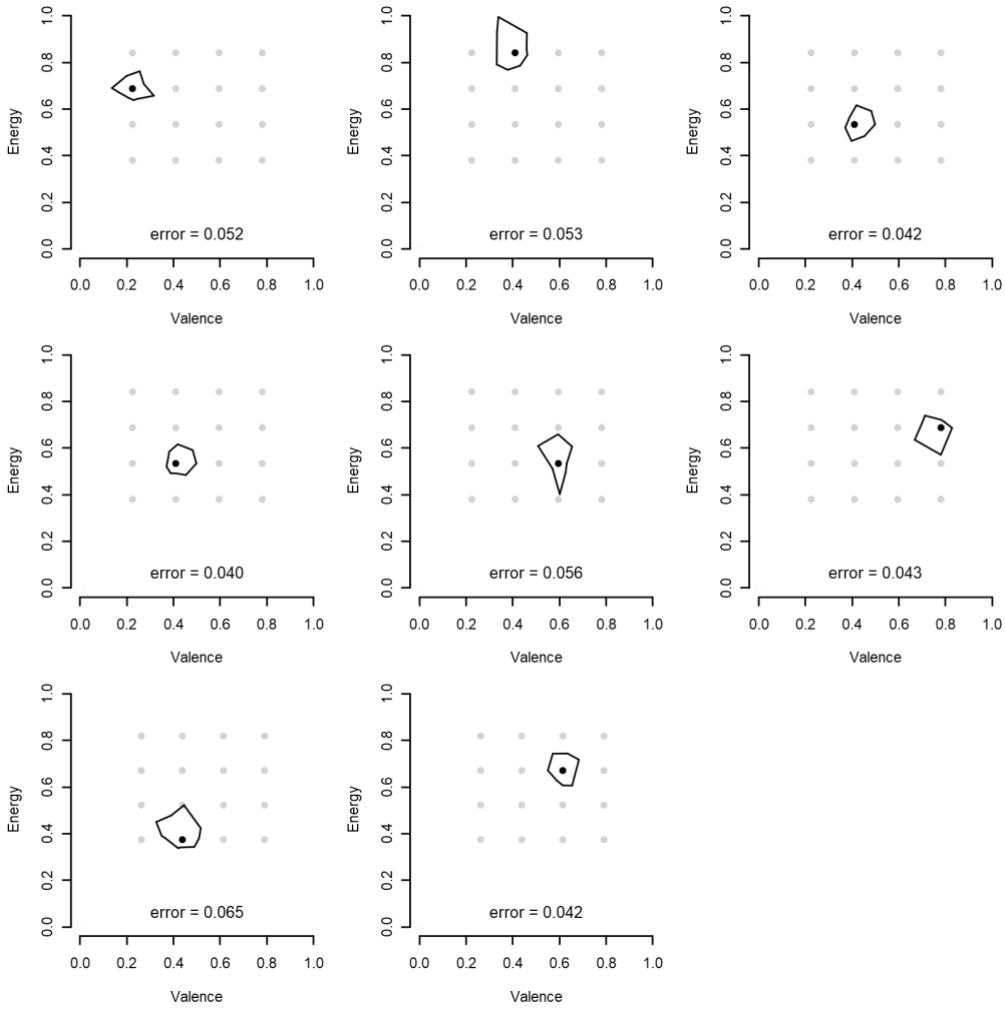
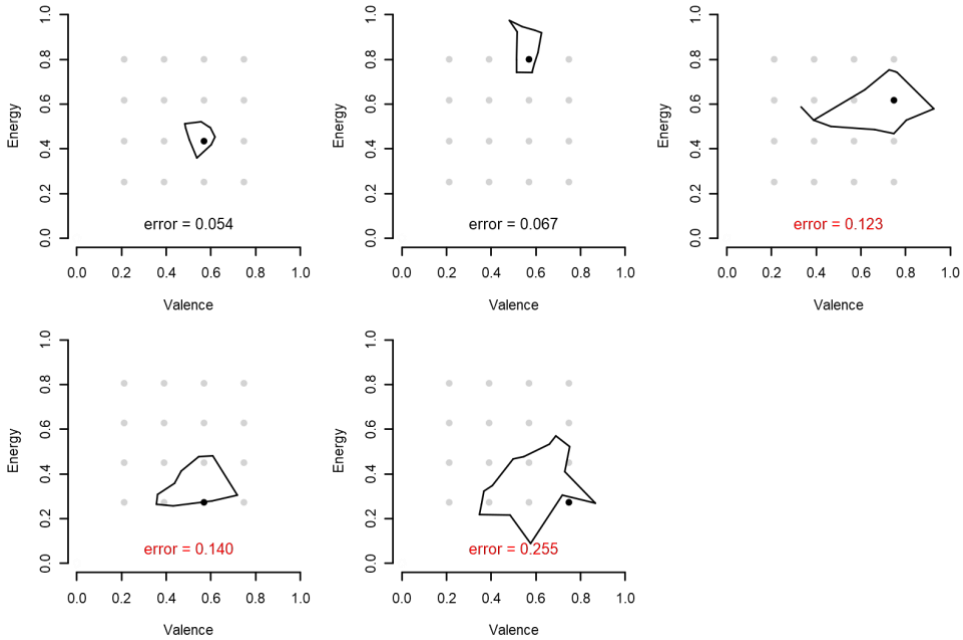
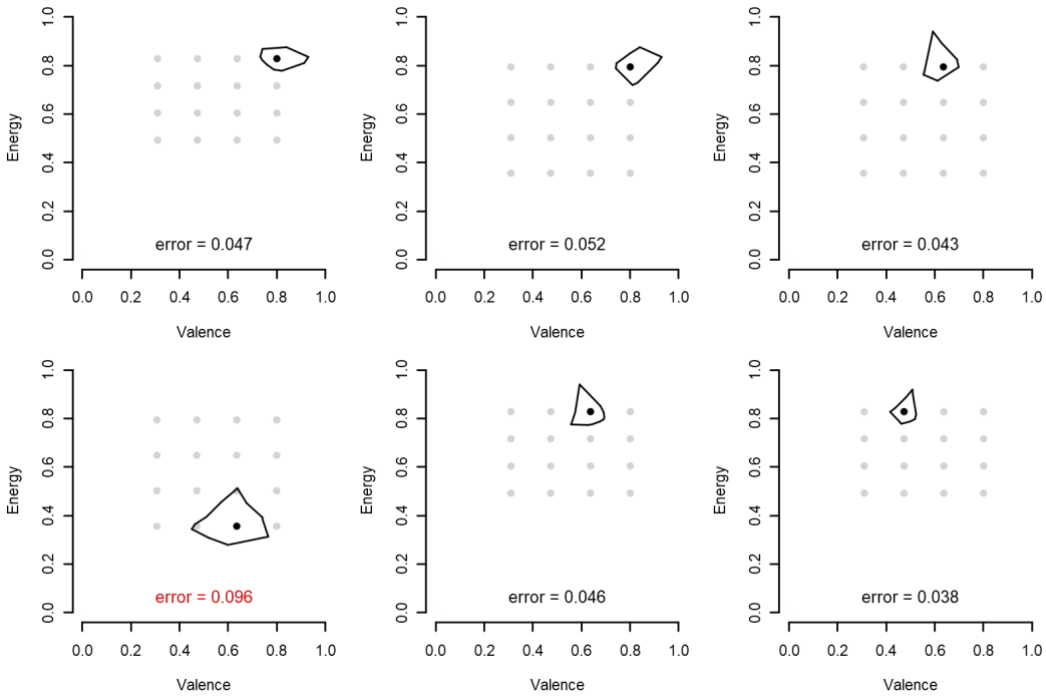


G.7 Participant 7



G.8 Participant 8



H Finishing interview answers

H.1 Listening patterns

Participant	How often did you use the application during the two weeks of the study?	How often did you make playlists?	How often did you listen to playlists?
1	Not that often. Don't listen to music that often, two songs walking to school. Kind of a barrier to leave Spotify, so not feel necessary to leave Spotify and find new songs just for those two songs.	About one, two a week maybe.	I listened to every playlist 2 times, so maybe 4-6 times overall.
2	Not very much, haven't counted. Listened a lot in the beginning.	Like 5-6 times, like only at the beginning, then I listened to them.	In the beginning it was every or every other day. At the end I haven't use it at all.
3	Once or twice a week. Maybe even more.	Every time I needed them. Sometime when I was going to study, when I was going to work out. Once when I was going to sleep.	Every time I created them.
4	Like every few days.	Every few days.	I listened to them right after I created them.
5	6-8 playlists I think. Every two, three days. When I got in a weird mood.	Every third day.	Twice when I created them. Like once right after, then once two days later.
6	Pretty often. Didn't make new ones every day, but used the lists I had quite a bit.	When felt that I was in a different mood and felt that I wanted to. Every second or third day maybe.	Pretty often. Not every day, but maybe every other day then.
7	Not so much last week, only twice, but otherwise I used it once a day when I thought of it. Not as much as I thought, but I didn't listen to much music this week.	I made a new list like every other day.	Just right after, listened straight away and then made a new list instead. Didn't go in and look up a previous list once, they were just there.
8	Twice a week maybe, was the average.	Every time I used it, so the same.	Listened to them right after I made them. Then the next time I made one, I listened to it. So only listened to each one once, except for one happy playlist.

Participant	Can you describe your typical usage patterns? **For example, in daytime/evening, while studying/working, on weekday/weekend?**	How did the application impact your music listening habits during the two weeks of the study?	Did you find yourself listening to more music than usual, or exploring new genres or artists?	Did you listen to the generated playlists more often than your own playlists or other music? Or was it the other way around?

1	Don't listen to music at home, just podcasts and YouTube. Just when walking somewhere or working out, but haven't worked out much lately so not much listening.	Found song in one list that I haven't listened to in years, so that was nice. But the lists became quite similar as I have quite narrow music taste and listen to like one genre of music. So many songs you recognized.	No more than usual.	Probably listened to these more than my own, but it's about the same songs just through the playlists.
2	When studying, mostly when at school.	Wasn't just listening to own classic playlists, found some new ones that I liked as well.	No, not more than usual. Usually listen to a lot of music, but away on a trip these too weeks so wasn't much music then.	In the beginning more of these lists, then switched to my usual ones.
3	Wasn't a specific day. Based more on what I was doing, like studying, sleeping, exercising, reading.	Don't listen to music constantly, so my listening habits probably haven't changed. Mostly listen when I'm studying and working out, otherwise I mostly listen to podcasts.	No not really, it didn't really change.	I listened more to the generated ones. It's really hard to put together your own playlist.
4	During the day, at work. On a computer. Probably did one on the mobile, but it was mainly on the computer.	Mainly that you listened to more unknown songs. Then maybe that you got a new song, liked the album and went there and listened a bit. So reminded of certain songs.	No, it was about the same as usual.	More on my own.
5	Mostly when I doing something else. More in the evening, usually when I was cooking or doing the dishes, or driving.	Listened to more different stuff, fun that you got to try out more music. Spotify's recs are a little too different, this was better because it was based on mood. Fit in more.	No, not more than usual.	Both, like an equal amount.
6	Mostly in the morning or in the evening. When I got ready in the morning and when I cleaned or cooked in the evening.	I found new songs that were good. A little more fun to listen to these because there were songs I had never listened to before, which I could then add to my own lists.	No more than usual, you kind of don't have time for that. I listened when I usually listen.	More on the generated ones, yes.
7	Mostly in the evening, if wanting a certain mood or atmosphere. Like when I was cooking, or studying, or in a certain mood and didn't know what else to listen to.	Fun because there was a lot of artists I hadn't listened to before. If I liked a certain song, I went in and listened to that artist instead of that list.	Yeah, I've started listening to some different, new artists during these weeks. And started to think more that I want to listen to a certain type of music based on what mood you're in.	About 50/50.
8	Mostly when working out at the gym. Once when I started cooking. Then once when I was walking home from school angry because something didn't go well.	I found some new artists, but not like a new genre. It was about the same genre as usual, that was the point.	Probably listened about the same amount.	More of these than my usual ones. Don't have a regular gym list so it was good to use this to create one.

H.2 Generated playlists

Participant	Did you find the personalized playlists generated by the application to be accurate and reflective of your music taste?	Were there any playlists that you found particularly accurate or inaccurate?
1	Yes, the playlists I made had songs I would listen to. There were just songs I recognized.	If I choose happy, I wouldn't call many of those songs since my songs are not that happy. Like violent songs. But I guess it's the happiest of my music taste.
2	Yes. But it was weird -- the "sad" list just became lo-fi. Accurate for my music taste, but I listen to other type of music too. But I guess it's what I listen to the most.	When I wanted calm, I got calm, so in that way very accurate. But when I think "sad" I think sad lyrics, not just the song. "Happy" was quite mixed, high tempo. All but "Angry" were kind of accurate.
3	Yes, very accurate. I think so.	Didn't think about it. Don't have good grasp of which songs I already have, so didn't reflect on if they were accurate. But probably thought it was a good mix of my own songs and new songs.
4	The majority of the songs did, that you had saved beforehand in other playlists. Then there were some outliers that stuck out. If you made a calm playlist, even if the song had a calmer tempo, the lyrics were more aggressive. Didn't always take the message of the song into consideration.	Liked my happy playlist, suited the mood. When you listen to music in the same session, its kind of the same type of songs. Like all Swedish classics in one go, but in that one it was mixed with more hyped songs. So it stuck out more but was still nice. So songs that you would never put in the same playlist yourself.
5	Yes, they reflected my taste in music. But sometimes not the mood.	Yeah, felt like 50/50, either it was on point or it didn't really know how to filter those songs.
6	Yes, I think so. There were songs that I hadn't listened to in years, they somehow made it in.	These "sad" lists were probably the best, depending on what kind of music you wanted of course.
7	The songs were good, but it was a little off on whether they matched the mood. It was different depending on whether you chose low or high. Maybe it didn't find music that fit.	Once I chose a "Happy" one and got a depressing list, so it didn't fit very well.
8	One had a lot of Norwegian songs and I don't really know why, but then I made a new one. Kind of matched the up tempo I wanted, since I was at the gym. All of them matched my music, except the Norwegian ones.	Happy, high energy was probably the one that was most similar to my taste in music and what I expected it to be.

Participant	Were there any genres, artists or individual songs that you felt were missing from the playlists?	Were there any specific genres or artists that you would like the application to exclude?	How important is it for you to have a wide variety of artists and genres represented in your playlists?

1	No, or there is probably some artist or sound that you come up with that doesn't make it, but then you haven't listened to it for a long time. But overall I'd say it fits well.	No. Maybe there could have been more sorting. Like when choosing angry, I would like more hard rock and not just rap, so maybe they could have set it in that direction.	Pretty important. Likes to listen many different artists, it gets pretty boring if a playlist is just an entire album.
2	Yes, on my "Sad" it was only lo-fi. That's what I listen to, but I would have liked a bit more artists that sing about sad stories and such. I don't get sad from lo-fi, I get calm. And in "Angry", the artists were not so angry.	Not that I can think of.	Not that important. Good music is good music. Can be only one artist who has a lot of good songs.
3	No. Not from my own perspective.	The genres and songs were pretty accurate based on my music taste. Have a mixed music taste, so it's difficult to know what to exclude. Wasn't like some random song cam in, that stood out.	Not really important. Don't usually explore other genres, have my own style. I get stuck a lot - If I find a genre I like, I stick with it for years.
4	Because it's so mixed up and much going on, you don't really think about what's missing.	Thought it was nicer to have more songs that you had heard before. In theory, it would have been nice to get recommended songs but once you sit there, you sort of skip them.	In one list it should be the same vibe, similar artists and songs. It doesn't have to be same genre but that it gives me the same feeling, personally.
5	No, not that I know of. Didn't know those out of place either. Might have wanted other songs that fit in from the artist that was included instead.	The depressing songs. I kind of brushed past them, because they didn't fit. But not exactly an artist or something, that I want to remove some just because they were new.	Pretty important, because every list I have is not just one artist. It's a mix of everything apparently.
6	No, I can't think of anything.	No, actually not. Not that I can think of. It was good music.	Very important, otherwise you get very bored listening to the same thing all the time. so it was very good to get such a variety.
7	Probably more that there were many lists that weren't that upbeat. Would like some cheerful lists.	Possibly some but can't say specifically. The first generated lists were just an artist I listened to constantly the week before, so it wasn't that broad. Maybe more variety there.	Very important.
8	Yes, I listen to Twenty One Pilots quite a bit but there weren't any of those songs. There were also not many artists I listen to, mostly new artists.	Norwegian music then. Otherwise, no.	Very important beacuse I like to find new ones. When I like an artist, I can kind of just listen to that one, but it's good to find new ones too.

Participant	**How familiar did you find the songs and artists in your generated playlists to be?**	**Playlists may have included songs that were not in your recent listening history. How appropriate did you find the inclusion of these songs?**
1	Recognized most of them. Around 20% new songs on each lists, but matched the others. But something new, so that's nice.	Thought they were appropiate, same vibe.

2	On a scale of 1-5, like 4. Recognized the majority. Most of them I listened to, then there were some I saw on the top charts and some were artists who were similar to artists you listened to.	In "Sad" they fit well, but not so well in the "Angry" playlist - no song that was angry there. Maybe I don't listen to angry music.
3	Thought it was a wide variety, a mix. Recognized some of the songs, but there were also songs that were completely new and fit right in.	Very fitting.
4	Recognized most of the artists, even though I don't actively listen to them. Many of the songs were also previously listened to, so it felt like they were pulled from my library.	The ones you recognized were appropriate but the others you thought "this doesn't fit", so you compared them. You hardly felt that you even wanted to listen to them or gave them opportunity to feel whether they fit in, judged them before you listened to them because the ones you knew stood out.
5	About 60-80% that I recognized. More of what I recognized, but also some new. Familiar in a sense that it's similar to what I've listened to as well.	It was appropriate. They fit in, they didn't stand out. Except for the ones that didn't fit in because they didn't match the genre that I thought matched the mood I had chosen.
6	Most of them I had somehow listened to before, or many years ago, but there were still many songs I had never listened to before. Or noticed.	I really thought they were great. It was fun to get something new.
7	A lot that was new, maybe only 2 songs in each that I had heard before. Still felt based on songs I had heard, so it matched the style. Still felt familiar, but was still somewhat new.	Yes, in general they were well generated.
8	Fairly familiar. There weren't many that you recognized, but a few in each playlist. So it was pretty much new, and not much familiar then.	Very. They fit in with what I did and I created the playlist for what I was going to do, the mood I wanted to do.

Participant	**Did you discover any new songs or artists through the playlists generated by the application?**	**How did you feel about these discoveries?**	**Did you add any of these songs or artists to your personal music library or playlists?**
1	Yes, I did.	Always nice. I like to broaden my music taste. Often you have a genre you like, then similar songs come along, then you like it too.	Yes, a few songs.
2	Yes.	It was fun. I was happy when I found something new, like these I can add to my personal playlist.	A couple, can't remember exact number.

3	Yes.	Good. Fun. Don't think it's fun to look for new artists and songs, just want them to be there when I want them. Nice not to have to search and find artists yourself and thus what you want to listen to.	No, I haven't, as I half-ignored my own playlists and only listened to the ones I had made. But if I were to do it, then yes.
4	Wouldn't say discovered, but rediscovered songs I interacted with before. Like I haven't listened to this in a while, let's bring it back.	No matter the mood, if you get a song you've listened to before you feel like "oooh" and grind them again. I can listen to the same song for like an hour. Then when you tire of them, and then find them again you grind them again.	The ones I recognized I had already saved somewhere, either in an album or playlist.
5	Yes. Two.	It felt good.	Yes.
6	I kind of don't have my own list so I don't notice the artists, but the songs I listened to were good.	Very good. You get a little more variety in what you listen to, you get tired of what you listen to after a while so it can be fun with new songs and artists.	No, I never use my own lists. I have old lists, so, but I never go in and listen. I listen more to the radio function, because I want something mixed.
7	Yes, I did.	Cool. It's nice not having to actively find artists elsewhere. You broaden your horizons a little.	Yes.
8	A lot of new songs by artists where I had just listened to one song before, but now they were many more by that particular artist. Which is incredibly fun, then I get to listen to other songs. Otherwise I don't know, didn't notice any specific artists that were better than the others.	Fun.	Yes, added one of them. NF had released a new song that I didn't know about so I added it. Also a french song that I liked on my "calm" list, which made me search for french music on Spotify.

H.3 Playlists vs. chosen mood

Participant	**Over the course of the study, how reflective did you find the generated playlists to be of your desired mood?**	**Were there any instances where the application generated a playlist that was not reflective of your desired mood?**	**How did you feel about the playlist in this situation?**	**Did you ever experience a change in mood while or after listening to a generated playlists?**

1	Fairly good. Probably hard with the music genre I listen to, so even though they're happy songs, the plot is not that happy. But otherwise it fits quite well.	No, not that I can think of.	-	No. I don't know, I'm not that easily affected. Not much can change my attitude.
2	"Happy" made me happy. "Angry" didn't make me that angry, and "Sad" made me calmer.	Didn't turn out quite as I thought, at least for "Angry". Then I kind of think Death Metal. But "Happy" was happy, would probably say it fits best.	No big deal, there was good music. I listened to it anyway, it was still good songs.	"Sad" calmed me down, so just wasn't the right mood. Should have another check that checks the lyrics in songs, if it is about something sad. I think listening to the happy playlist made me happier.
3	It varied a bit I think. Sometimes they were very accurate, and sometimes it was like this "but this song wasn't sad". But it wasn't like every song didn't fit in, in that case there were only a few that stood out.	No. Not a full playlist.	-	Fell asleep once, that was nice. At the gym, I got hyped or focused on what I was doing like I wanted. I don't know about changed mood, but I got to the mood I wanted to be in.
4	Pretty well when the message of the song matched the mood of the playlist. What I said before with the lyrics, when the message of the song is completely different. Mostly "Calm", didn't fit then. If you want to get "calm", you have to think of the song's message. If it's a calm song that is aggressive, then you don't think it's calm even though musically it is.	Wouldn't say whole playlists, more specific parts of it. "Sad" had some songs with pep vibe, even though they are calmer or slower. For "happy" you don't think as much about the lyrics, the important thing is that the song gives a good feeling -- musicality is the important thing. "Angry" -- take the genre into consideration instead of the tempo, e.g. that I want rap. Maybe to decide that yourself.	You didn't stay so long on it, you focused on the songs you recognized and wanted to listen to. And go to their song radios and find and queue similar ones that you knew.	No. I wouldn't say that. However, I've been in a rather wobbly mood these past few weeks, but still.
5	Pretty accurate, except for those occasional songs that reflected the opposite. It might also be that I listen to depressing songs when I'm happy, because they are nostalgic songs. Maybe it's me.	Yes, once I got a playlist that wasn't right, it was pretty bland. Didn't reflect what I had chosen at all. It was probably somewhere in the middle of happy and calm, but still got angry music.	It was a bit of a shame, but it was just a matter of making a new list and you could get a better one.	Yes. Most of the time I got better, I mostly made happy and calm playlists and then I got in that mood, it became a little more fun to do stuff. That was what I did the most.
6	It was very fitting. They were always songs I liked and they fit the mood I had chosen.	Probably "Angry", where there were Swedish songs that didn't fit in. One "Angry" was great and so accurate, but the other... So strange that it's so different. I clicked the same box.	Then I changed to "Angry #1" instead, because then I knew it was right.	I don't change my mood that much. Most of the time take a mood and you keep that mood, e.g. party mood -- keep the partyfeeling.

7	Some moods matched better, based on what I previously listened to. Sometimes it was very good, but some lists did not match at all. But I could just make a new list then.	Yes, when I made an "Angry" playlist it was weird - not bad - but didn't match my mood, what I would say is angry music. When I made a "calm" list it was very depressing. It was either slow and depressing or the complete opposite.	Listened to the first playlist a bit anyway, because there were many new songs, but it wasn't the mood I wanted. Was a bit off to listen to, because I knew what I wanted and it didn't match at all.	Yes. I mostly made lists that I thought matched my current mood, so it was more lists that reinforced my current mood instead of completely changing it.
8	Mostly chose high energy and happy and they all had that tone. Chose some calm and some angry list, and the angry one had angry text and the calm was mostly music and no text - which was pretty calm. Think they matched that mood pretty well and what I expected when I chose that mood.	No. Not that I made a lot of playlists, but the ones I made were accurate.	-	Yes, kind of. Don't know if it was from being in the gym or from the playlist. But mostly yes, I was happier after listening to the playlist. BUT a calm playlist actually made me very calm. It was very good.

H.4 Using MOODsic

Participant	**Did you find the user interface of the application to be user-friendly and easy to navigate?**	**Were there any particular aspects of the interface that you found useful or challenging?**	**Were there any features that you found difficult to use or understand?**	**How did you overcome these challenges?**
1	Every time I use it, I need to connect to Spotify -- bit of a hurdle. Otherwise it makes sense, not much content. Can clearly see what mood to choose, so it feels good.	That it is a web service, if it had been an app it would have helped. Would have felt better as an app, like Shazam. If integrated with Spotify, probably would have used it more.	No. You choose a mood then get a playlist, not much to understand.	-
2	Very user friendly. No problem to find. Found where I wanted.	Nothing difficult. It was very easy to generate a playlist. I had no problem at all.	No, if I haven't missed something. I was able to login, generate playlist. The graph was very clear as well. The colors also fit, they felt right. It was reasonable.	-
3	Yes. You don't need to interact with it that much to generate a playlist. There weren't that many steps to fill in to get something.	Mostly used the standard buttons, they work well. You understand everything, wasn't hard to find things. So don't think so.	Probably at first that you had to think when choosing based on the color and the diagram. You had to think what was what. But then after making a few playlists you learned.	You just had to think logically.

4	Yes, mainly because it was pretty clear what to do. It wasn't 15 steps, just one question. And even though boxes were different buttons, they still felt connected so you had to choose on. It wasn't step by step, just one mission. I also like that it was interactive, like hover, that was nice. Not difficult interpreting the different axes.	Good that the button says what I should do in the next step, so I can focus only on picking mood. It was useful. Then you get to the page with "you want..", I liked that. Same with "generate playlist", then I know what will happen next, something will happen. Should I regret it, I feel like I can back out. Same with "how do you want to feel", you get a confirmation beforehand. And the confirmation page was nice, "pick new mood" etc. I liked it.	No, it is actually very simple to use.	-
5	Yes. There's not much to misclick there. I used it mostly on the phone.	Text on the cubes that say what kind of mood it is. It can be difficult to navigate if you don't slide to the side and see what it is, but it wasn't something that hindered me directly. Difficult if you are color blind too.	No. Maybe would have liked the mood picker right away, but I understand why the start page is first.	I clicked the button.
6	Yes, I think so. It was very good. It looked professional and well organized.	No, I can't think of anything.	No. Maybe what box to pick, because it was just "a little less angry". But you could still figure that out.	You understand, you learn after a while.
7	Yes I think so. It worked well.	At first a bit difficult to interpret the graph and hard to predict exactly what mood it would be. Didn't know how to think at first, but then there was no problem.	The graph, in that case.	A bit of testing in the beginning, the first playlist was more tricky. But when the playlist is created, you see that it is "happy" so then you get a picture of how it works.
8	Yes. I was already logged in to Spotify, so it was incredibly nice. I liked that the mood graph was visible on the whole screen, didn't have to scroll. Didn't notice the menu in mobile mood.	Yes, liked that you can open the generated one in Spotify. Hardly used the "pick a new mood" button -- once you've created one, you don't want to create a new one.	No, not at all. There's not a lot of stuff you do, it's pretty straightforward.	-

Participant	**Were there any features that you think would be useful but were not included?**	**Would you continue to use the application in your everyday life beyond the study period? Why or why not?**	**Can you describe how you might incorporate the application into your daily routine?**

1	Integrate with Spotify. Maybe that you can control a little more which genre you want to get. Because even if you choose sad and still get rap, you might want to choose Lewis Capaldi instead.	Probably not. If it had been part of Spotify, I would have used it, but having to go into another service is too much work for the little I listen to.	Every time you go in, you're in the mood for new music, you make a new list. But if it had been in Spotify, I wouldn't have wanted it to be saved every time, and instead only generated once. That way you avoid getting a long list of playlists, then only one will be updated.
2	Get it to not go through your own history, but just take some sad songs, nothing specific, just a mix. It would have been a fun thing. If you feel like exploring a bit, kind of.	It is very possible. Spotify has the ability to generate a playlist based on a song, so it competes a bit. But yes, easy. I would probably make a playlist again. To find new songs in that case, so you take the best from the newly generated playlists and put in your own.	Maybe when I'm sitting at home and going to play and listen to some music, maybe I feel like listening to some new songs, then I'd probably make a new list. Then maybe you make a new one when you've listened through it.
3	That you had to design your own front image. Or choose between how it would look, like an avatar depending on the mood. Like choose a smiley that was happy if it's a happy playlist.	Yes. 100%. Because I think it's a very good idea and I think it's hard to look for new artists and songs that fit my style of music, then you don't have to search Spotify for hours and fast forward songs to see if they're good or not.	Well, I don't think I would have used it every day, but I probably would have when I listened through a list and got tired of it. But maybe every two weeks to get new songs and not listen to the same old regular ones.
4	What if you don't want recommended songs but only want songs you've listened to before, like if you have the feeling "I don't like new songs" but only want songs in your own library and want a mixture of it. Then also… you still have the feeling that there are more songs in those 1.5 hours. So maybe you could choose the length of the playlist. And if you could sort of center it around a genre sort of, combine both mood and sort of rock songs. Sometime when I have high energy and high happiness I might want to listen to specific bubblegum pop.	I could imagine using it if there were more choices before the list was created. To be able to control the content a little more yourself. That is the whole charm of having your own playlist, that you have curated it yourself and here you get a little help along the way, but you still want to feel that you have contributed to it yourself.	If I had to choose completely myself as mentioned before, I would have liked a button in Spotify, like "whats your mood today" and from there expand the steps that exist and include the options I just said. But that it is integrated so that it becomes a smoother experience. Or that you get some kind of reminder to do, like in the morning when you start Spotify for the first time of the day. Like "Hey how are you today, how do you want to feel today?", kind of like that. That it pops up a little. But then it might feel disturbing. But maybe to control when it pops up. Or that Spotify notices that now she is wavering a bit. Or face recognition, "Now she looks a bit bitter, shall we go and help her a bit?".
5	In an app would have been nice. That's the only thing that could have been better then. Voice control too, that would have been nice. But it is a bit difficult to do. Maybe connect it to Google, so you can say “Hey Google”, I want to listen to this and get in this mood.	Yes. Because it worked well. It did its job, it made me happier when I made a happy playlist. And it was easy to use.	If I'm cooking or something, when I'm doing the dishes. Not when I'm going out for a run maybe, I haven't tested it so I don't know how effective it is. But when I'm doing something else at the same time, probably.

6	No, I can't think of anything that could have been changed. Because it is saved regardless of whether you clicked into Spotify or not, so no.	Yes, I actually think so. I would have liked it to be there when you want to get a little more music or inspiration. Because then you get new songs you've never heard before, so it won't be the same old ones you've always had. And that you can choose which mood you want to be in and get a playlist based on that.	I would probably have kept the lists I had already created until I feel that "now I want a new one". So I don't think I would have created new lists every day, but only when I got tired of an old one and want to create a new one.